

DYNAMICMS SUITE

Dynamic Apps Standard SRS

Common Software Requirements Specification for DynamicMS Suite Applications

Consolidated from journi (DynamicJourni) v2.0 and NCP Solver v2.0, extended from the CortexPLM reference build

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) — “Dynamic Apps Standard SRS” — defines the set of functional and non-functional requirements that are common to every application in the DynamicMS Suite, irrespective of the business domain a given application addresses. It is derived by comparing the individual SRS documents of two suite members — *journi* (DynamicJourni), a human change-management application, and NCP Solver, a non-conformity and problem-resolution application — and extracting the requirements, architecture patterns, and governance mechanisms that recur identically, or near-identically, across both.

This document is intended to serve as the baseline specification a new Dynamic App is built against, and as the reference against which an existing Dynamic App's platform layer (as distinct from its domain-specific modules) is assessed for conformance. It is not a replacement for either product-specific SRS: *journi* and NCP Solver each retain their own SRS for domain-specific functional requirements (the 22 change-management modules on one side, the S1–S7 NCP Sheet workflow on the other).

Revision 1.1 extends the standard with platform capabilities first built and verified in CortexPLM, an innovation-lifecycle Dynamic App, where they proved independent of that application's business domain — for example the pinnable, repositionable navigation menu, internal benchmarking, and in-place upgrade and backup. Only requirements applicable to any Dynamic App were retained; CortexPLM's phase-gate lifecycle and its product-specific content remain in its own documentation. Appendix B lists every change made in this revision.

Revision 1.2 adds a second set of capabilities proven in CortexPLM that any Dynamic App can reuse: the tenancy tree (Group Yes/No > Organization > principal records) and the provisioning of new organizations, the portfolio overview matrix, a stage checklist library, multi-format attachments, AI shown and run where it applies in the process design, the choice of the live AI model, a full-screen BPMN workspace, on-screen legends and a decision matrix for scored fields. Appendix C lists these changes.

Revision 1.3 integrates the application-agnostic requirements for the Vertical and SME layers — the vertical framework, vertical processes, data, compliance, integration and configuration, the SME mode with its tracks, complexity scoring, onboarding, packaging and pricing, and their combination — restated as Dynamic App requirements. It also adds, for any Dynamic App, a project template catalog (universal or per vertical, Full or SME mode) with full CRUD, three ways to create a project (from the catalog, manually, or with AI), a Checklist Library whose checklists may or may not be related to a gate, and a Gate Library to which checklists are associated; both are managed with full CRUD and can be attached to any project phase. Appendix D lists the changes and Appendix E traces every source requirement family to its requirements here.

Revision 1.4 adds standards and requirements management for any Dynamic App: a Standards Catalog holding internal frameworks and external standards (ISO, IEC, regulations, industry and customer standards) structured from part and chapter down to requirement; a Requirements Catalog in which every requirement, whatever its origin, is one typed record; and the applicability of a whole standard or of individual requirements to an Organization and/or a project, with tailoring, compliance status, evidence and a statement of applicability. Appendix F lists the changes.

1.2 Scope

In scope: the tenant and organizational hierarchy; identity and role-based access control; Configuration Management (Solution Packs, Add-Ons, Compliance & Security Standards); the governed AI Use Case Library; per-process/per-stage governance artifacts (Alerts, Business Rules, COSO-classified Controls, Risks & Opportunities, Custom KPIs, RACSI accountability); a Standards/Knowledge Base with RAG grounding; BPMN 2.0 process modeling; standard reports and data export; an in-app Notification Center; an AI Assistant / chatbot offering data-query and application-help modes; an audited, justification-governed change log; an in-app Help guide; template libraries; multi-language localization including Arabic RTL; and the non-functional requirements (performance, security, reliability, usability, portability, maintainability, compatibility, data retention) these capabilities are held to.

Revision 1.1 adds to this scope: a personalizable navigation shell (a menu pinned open or sliding away, docked on any edge of the screen, with favorites that follow the user); internal benchmarking within an Organization and across the Organizations of one Group; and platform operations (scheduled backup with retention, additive startup migrations, startup guards, a health endpoint, a reproducible demonstration seed, and continuous integration across operating systems).

Out of scope: any requirement specific to a single application's business domain — journi's 22 change-management modules and their ADKAR/Lewin/Bridges/Kotter/Kübler-Ross framework content, or NCP Solver's seven-stage NCP Sheet lifecycle and its Ishikawa/5-Why root-cause tooling. Those remain the authority of their respective product SRS documents.

Also out of scope: external benchmarking — any comparison with an organization outside the requester's Group, or with third-party or industry data.

1.3 Intended Audience

Product owners defining a new Dynamic App; architects and developers building or rebuilding a Dynamic App's platform layer; quality assurance engineers verifying platform-level conformance; and auditors assessing governance, security, and compliance posture across the suite.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
Dynamic App	Any application built on the DynamicMS Suite platform pattern described in this document (e.g. journi / DynamicJourni, NCP Solver).
Solution Pack	One of three commercial tiers defining an Organization's feature and quota entitlement matrix.
Add-On	A priced, independently toggleable capability that grants a feature to a Pack not already including it standard.
Compliance & Security Standard	An independently priced/toggleable module (e.g. GDPR, ISO/IEC 27001, SOC 2) that seeds GRC scaffolding; explicitly not a certification.
AI Use Case	A governed, versioned catalog entry describing one AI-assisted capability, tiered Assistive or Augmented (never Autonomous).
RAG	Retrieval-Augmented Generation — search over a knowledge corpus (TF-IDF/cosine-similarity or equivalent) grounding an AI suggestion or chatbot answer in the tenant's own data or a static reference corpus, without requiring a live external LLM call.
OBS	Organizational Breakdown Structure — the site/department/team or resourcing tree other

Term	Definition
	governance modules reference for ownership and assignment.
RACSI	Responsible, Accountable, Consulted, Support, Informed — a five-role accountability model; exactly one Accountable per activity.
COSO	Committee of Sponsoring Organizations of the Treadway Commission — the internal-control framework used to classify Controls.
BPMN	Business Process Model and Notation — the OMG 2.0 standard used to model process workflows in-browser.
RBAC	Role-Based Access Control — access rights determined by one or more assigned roles, mediated by a runtime-editable Permission Matrix.
Justification Governance	The stage-then-justify pattern: a practitioner stages a new value, records a justification note, and only then commits the change, appended atomically to an audit/change log.
Multi-Tenant	Architecture in which multiple Organizations share one deployment with isolated data.
Work Breakdown Structure (WBS)	A hierarchical decomposition of any grouping of Actions into a tree of WBS nodes, used as the basis for Gantt scheduling.
Gantt Chart	A timeline rendering of a WBS, plotting each node/Action's planned dates and dependencies.
Version Management	The platform's generic service for versioning any registered entity type: immutable history, no destructive edits, and revert-by-duplication to a new Current version.
Multi-Channel Communication	Delivery of a notification or alert through one or more channels (in-app, email, SMS, push, webhook) behind a single shared dispatch interface.
Mobile Compatibility	A responsive layout, meeting WCAG 2.1 AA touch-target sizing, that keeps an application's core workflows fully usable on a mobile viewport without a separate native application.
Return on Experience (REX)	A structured lessons-learned entry captured at the closure of a Project or an Action — what went well, what didn't, root cause, recommendation — indexed for future retrieval.
External Integration	A registered, independently toggleable connection from the platform to an external application, configured through the External Integration Registry (Section 3.3).
Navigation Shell	The frame shared by every authenticated screen: the primary menu, the header bar (context switcher, language, notifications, AI Assistant, user menu), and the content area.
Pinned / Sliding Menu	The two menu modes. Pinned, the menu stays open and the content reflows beside it. Unpinned, it collapses to an edge handle and slides over the content when pointed at, focused, or toggled.
Dock Position	The screen edge — left, right, top, or bottom — a user chooses for the primary menu; a per-user preference stored server-side.
Internal Benchmarking	Comparison of standard metrics between segments of one Organization, or between Organizations of the same Group, using aggregates only. External benchmarking is out of scope.
Minimum Sample Size	The smallest number of records a benchmark segment or Organization must contain to be ranked; smaller samples are shown as not comparable.
Additive Migration	A change applied at startup that only adds tables, columns, permission codes, or default grants, and never drops or rewrites existing data or an administrator's edits.
Tenancy Tree	The view Group (Yes/No) > Organization > principal records, limited to what the viewer may see; an Organization with no Group appears as Independent.
Provisioning	Everything a new Organization receives on creation: OBS skeleton, standard governance catalog, AI Use Case Library, checklist and record templates, and optionally a starting team.
Portfolio Overview	A matrix of principal records (rows) against process stages (columns) whose cells show the status of each stage, for a chosen scope.
Checklist Template	A reusable, versioned list of checklist items kept in the Checklist Library. It stands on its

Term	Definition
	own and may be associated with zero, one or several gates of the Gate Library; when associated, it is copied into the gate checklist when the phase opens.
Vertical	An industry sector (for example Automotive, Food & Beverage, Public Education) defined as a first-class, versioned configuration entity that activates its own macro processes, E2E processes, data extensions, compliance mappings, checklists and gate criteria.
Vertical Instance	The content of one vertical — its prefix, macro and E2E processes, data model, compliance, integration, configuration and governance — delivered as configuration data, not as platform code.
SME	A small or medium-sized enterprise, by default 1–250 employees (EU) or up to 500 (US SBA), thresholds configurable per region, segmented into Micro (1–10), Small (10–50) and Medium (50–250).
Mode (Full / SME)	The rigor profile of a project: Full mode runs the complete process set; SME mode runs a right-sized SME track.
SME Track	One of two to four configurable levels of process rigor for SME mode, each defining its macro and E2E processes, gates, checklist items per gate and typical duration.
Complexity Score	A weighted score of configurable criteria (weights summing to 100 %), including vertical-specific drivers, that recommends the track of a project.
Project Template Catalog	The governed, versioned library of project templates, each classified by scope (universal or one vertical) and mode (Full or SME, with its track).
Creation Mode	The way a project is created: from the catalog, manually, or with AI (an AI-drafted project reviewed and accepted by a person).
Phase	A stage of a project's process chain (for example an E2E process or a lifecycle stage) to which gates and checklists are attached.
Gate Definition	A reusable decision point — name, criteria, approvers, decision options Go / No-Go / Hold / Recycle — kept in the Gate Library, with zero or more associated checklists, and attachable to any phase.
Pack / Bundle	A priced commercial package: an SME pack, a vertical pack, or a bundle of both.
Standard	A body of requirements kept in the Standards Catalog, either Internal (an Organization's own framework, policy or method) or External (an international, national or industry standard, a regulation, or a customer standard), identified by code, title, issuer and edition.
Standard Structure	The ordered tree of a standard — part, chapter, clause, sub-clause to any depth — whose leaves are requirements.
Requirement	A single, verifiable statement kept once in the Requirements Catalog, with a type, an obligation level (shall, should, may), a verification method and an owner; it may belong to a standard or stand alone.
Requirement Type	A configurable classification of requirements by origin or nature, for example Standard, Internal, Regulatory, Contractual, Product, Process, Security & Privacy, Quality.
Applicability	The link that makes a standard, a clause or a requirement apply to an Organization and/or a project, with its status: Applicable, Not applicable (justified) or Tailored.
Statement of Applicability	The list of every requirement in scope for an Organization or a project, with its applicability, justification, compliance status and evidence.
Compliance Status	The assessed state of an applicable requirement: Not assessed, Compliant, Partially compliant, Non-compliant, or Not applicable.

1.5 References

- [journi](#) — Software Requirements Specification, Version 2.0, September 2026.
- [NCP Solver](#) — DynamicMS Suite Software Requirements Specification, Version 2.0, September 2026.

- Dynamic Apps Suite — Brand & Graphical Chart Guide, Version 1.0, August 2026.
- CortexPLM — Reference implementation of the Dynamic Apps platform layer, build of September 2026, with its SRS Gap Analysis, September 2026.
- Application-Agnostic Requirements for Verticals and SME — Vertical and SME layers of any PLM or PSLM platform (29 verticals, SME framework, Vertical $\times$ SME intersection, non-functional requirements, traceability and requirements governance), 2026.
- ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering.
- OMG Requirements Interchange Format (ReqIF), Version 1.2.

1.6 Document Overview

Section 2 describes the Dynamic Apps Suite at a high level and states the design principles every Dynamic App is expected to follow. Section 3 defines the common platform architecture. Section 4 enumerates the common functional requirements, grouped by shared capability and individually identified (FR-DA-xxx) for traceability. Section 5 defines common non-functional requirements (NFR-DA-xxx). Section 6 summarizes the common data-model entities. Section 7 defines common external interface conventions. Appendix A records where JOURNI and NCP Solver currently diverge from this standard, so the gap is explicit rather than silently absorbed.

Revision 1.1 adds Sections 3.11 to 3.13 and 4.19 to 4.21, adds requirements to existing sections under new, never-reused IDs, and adds Appendix B, the change log of this revision. No 1.0 requirement was removed or renumbered.

Revision 1.2 adds Sections 4.22 to 4.24, new requirements in Sections 4.1, 4.4, 4.7, 4.12, 4.19, 4.21 and 5.5, and Appendix C, the change log of revision 1.2. As in 1.1, nothing was removed or renumbered.

Revision 1.3 adds Section 3.14, Sections 4.25 to 4.35, new requirements in Sections 4.13, 5.1 to 5.6, 5.8 and 5.10, and Appendices D (change log) and E (traceability to the Verticals and SME source). It renames Section 4.23 Checklist Library and rewords FR-DA-CHK-01 to 05 so that a checklist no longer has to belong to a gate; Section 4.35, the Gate Library, holds the gates and their associated checklists. Nothing was removed or renumbered.

Revision 1.4 adds Section 3.15, Sections 4.36 to 4.38, new requirements in Sections 4.3, 4.5, 4.6, 4.23, 4.27, 5.1, 5.4, 5.9 and 5.10, and Appendix F, the change log of revision 1.4. Nothing was removed or renumbered.

2. Overall Description

2.1 Product Perspective

A Dynamic App is a self-contained, installable web application — a browser-based single-page frontend paired with a REST API backend — that can run standalone or, per NFR-DA-MAINT-01, expose its composable modules (Action/Task Register, Alert & Notification Engine, RBAC layer, KPI/Reporting engine, Audit Trail Service) for reuse by sibling Dynamic Apps behind a shared API contract. Every Dynamic App is positioned as data- and AI-led: its deliverables are evidence-based records, structured workflows, and governed AI-assisted suggestions, never autonomous decisions.

2.2 Design Principles

- Explainable AI only. Every AI capability is tiered Assistive or Augmented; a third, Autonomous tier — an AI that takes an irreversible action without a human decision in the loop — is out of scope for every Dynamic App.
- Tenant isolation by construction. Every data-bearing record is scoped to an Organization; no query, export, or AI retrieval may return another tenant's data.
- Commercial gating as a second, independent axis. Solution Pack and Add-On/Compliance entitlement gate feature and quota access alongside, never instead of, RBAC.
- Governance is structured, not narrative. Business rules, controls, risks, and accountability are captured as typed, queryable records — tagged back to a process or stage — rather than left in prose documents.
- One shared visual and interaction language. Every Dynamic App, and every document it generates, conforms to the suite's common visual identity (color palette, typography, iconography) across every module, output format, and supported language.
- Multilingual and RTL-first. English, French, and Arabic (full right-to-left layout) are first-class from the outset, not a later localization pass.
- One navigation shell, personal to each user. Every Dynamic App shares one navigation shell; each user may pin the menu or let it slide away, dock it on any screen edge, and mark favorites, and these choices follow the user across sessions and devices.
- Compare within the family, never outside it. Benchmarking compares an Organization with itself and with its own Group, from aggregates only; it never exposes another tenant's records and never reaches beyond the Group.
- Upgrade and recover in place. A new release upgrades an existing installation at startup without data loss, and the data is backed up on a schedule the operator does not have to remember.

2.3 Product Functions (Summary)

- Register, resolve/manage, and evidence a structured record type (an NCP Sheet, a Change Management Project) through a defined multi-stage lifecycle.
- Provide one or more rule-based AI agents grounded in a retrieval corpus, plus a chat-style AI Assistant offering RBAC-scoped data queries and application-help queries.
- Maintain a governed AI Use Case Library — a seeded catalog plus organization-authored custom entries, versioned, with Organization-level and Project-level activation.
- Maintain per-process/per-stage governance: Alerts, Business Rules, COSO-classified Controls, Risks & Opportunities, Custom KPIs, and a RACSI accountability matrix.
- Model process workflows as real BPMN 2.0 diagrams, importable and exportable as standard XML.

- Provide Configuration Management: Solution Packs, priced Add-Ons, Compliance & Security Standards, deployment options, and support tiers, each enforced wherever the platform gates a feature.
- Provide full role-based access control with a runtime-editable Permission Matrix, and an organizational hierarchy (Group/Organization/Project) with an OBS for people/department scoping.
- Compute standard KPIs plus Custom KPIs, and standard Reports exportable to PDF, Excel, and Word; support CSV export on high-traffic tables.
- Operate multi-tenant with data isolation, and multi-language with full Arabic RTL support.
- Provide a searchable, multi-language in-app Help guide covering every module.
- Provide a personalizable navigation shell: menu pinned or sliding, docked left, right, top, or bottom, with favorites, persisted per user.
- Provide internal benchmarking of standard metrics between segments of one Organization and between Organizations of the same Group.
- Protect and upgrade data in place: on-demand and daily backups with retention, and additive migrations at startup.
- Activate verticals and an SME mode by configuration, score project complexity to choose the right track, and start projects from a governed template catalog, by hand, or with AI, with gates and checklists attached to any phase.
- Manage internal and external standards from chapter to requirement, keep every requirement in one typed catalog, apply standards and requirements to Organizations and projects, and track compliance with evidence.

2.4 User Classes and Characteristics

Every Dynamic App defines its own domain-specific role names, but each role maps onto one of the following six baseline classes, which the common Permission Matrix (Section 4.2) is built around:

Baseline Class	Characteristics	Example Mapping
Platform / Organization Administrator	Full configuration authority: Configuration Management, RBAC, hierarchy, AI Use Case governance, Compliance Standards.	Administrator (NCP Solver); Super Admin, Group Admin, Organization Admin (journi)
Process Owner / Manager	Owns the governed process end-to-end: validates records, configures thresholds, owns Controls/Business Rules, full governance CRUD.	Quality Manager, CI Pilot (NCP Solver); Change Manager (journi)
Team Contributor	Creates/edits records and actions within an assigned scope; completes assigned workflow stages.	NCP Team Member (NCP Solver); Practitioner (journi)
Action Owner / Evaluator	Executes or verifies a specific assigned action; owner and evaluator of the same action must always differ.	Action Owner (AR) / Evaluator (AE) (NCP Solver); comparable action-level roles (journi)
Scoped Viewer / Sponsor	Department- or team-scoped visibility into records, dashboards, and governance items relevant to that scope.	Department Head (NCP Solver); Sponsor, People Manager (journi)
Reporter / Read-Only / Auditor	Either submission-only creation with no post-submission edit rights, or full read-only access including the audit trail.	Reporter, Read-Only/Auditor (NCP Solver);

Baseline Class	Characteristics	Example Mapping
		Employee, Executive (journi)

2.5 Operating Environment

- Server: a standard Node.js runtime (Node.js 18+ minimum; Node.js 22.13+ recommended — the first long-term-support line whose built-in SQLite module works without an experimental flag — so the platform can use the runtime's own SQLite module rather than a native-compiled dependency). The required version shall be declared in the package manifest.
- Client: any evergreen desktop, tablet, or mobile web browser, with a responsive layout adapting to each; no legacy-browser support is in scope.
- No mandatory external service dependency: a Dynamic App must remain fully functional with no live third-party call, with the sole opt-in exception of a user-configured Real LLM Provider Connection (Section 4.3).
- Report-export and persistence libraries must be pure JavaScript / native-runtime, so installation succeeds without a native C/C++ build toolchain, a required container orchestrator, or an external database server.

2.6 Design and Implementation Constraints

- Every data access must be scoped by tenant (organization_id); no query may return another tenant's data.
- All AI agent behavior must be explainable and logged; no black-box autonomous decision-making.
- The UI, and every generated document, must conform to the suite's common visual identity across every module, output format, and supported language.
- An action's Owner and its Evaluator, wherever the platform models a two-party verification step, must always be different users, enforced server-side.
- Solution Pack entitlement is a second, independent gating axis alongside RBAC, and every Pack quota must be checked server-side at the point of creation/activation.
- Every user-facing string must exist in a single translation dictionary for every supported language; no hardcoded UI string is acceptable outside it.
- Benchmark data leaving an Organization, even towards its own Group, must be aggregated; no record, name, or identifier of another tenant may be returned.

2.7 Assumptions and Dependencies

- Organizations are responsible for their own user provisioning within the tenant they administer.
- Rule-based AI agent implementations are a functional, explainable stand-in that a future LLM/vector-database-backed implementation can replace behind the same internal interface, without changing calling code.
- Compliance & Security Standard modules provide GRC scaffolding only; an Organization remains responsible for its own compliance program and any external certification.
- Demo/seed data is fictional and is not a substitute for real organizational data before go-live.
- Off-site copies of backups, and the hosting platform's own disaster-recovery arrangements, remain the responsibility of the operator of each deployment.

3. Common Platform Architecture

3.1 Tenancy & Hierarchy

Every Dynamic App is built on the same three-to-four-level hierarchy: an optional Group above the Organization (tenant) level, the Organization itself as the hard data-isolation boundary, and one or more Projects below it. An Organizational Breakdown Structure (OBS) — a site/department/service/team tree, or an equivalent per-project resourcing roster — supplies the shared 'who/where' backbone that governance modules (Business Rules, Controls, Risks & Opportunities, RACSI, BPMN diagrams) point back to.

Group membership is also the only boundary across which Organizations may compare performance (Section 3.12). Every hierarchy screen shows whether an Organization belongs to a Group (Yes/No) and which one; an Organization with no Group is shown as Independent.

3.2 Identity & Access Control

A Dynamic App recognizes a fixed catalog of standard roles (nine in the reference implementations) and supports assigning one or more roles per user. Every write/manage capability check reads from a runtime-editable Permission Matrix — a role × capability grid, seeded with sensible defaults, editable in place by a Platform/Organization Administrator, and effective immediately, platform-wide, without a code change or restart. The matrix is a fail-safe override on top of an underlying default model, never a bypass of it.

3.3 Configuration Management & Commercial Gating

A single Configuration Management surface records, per Organization, exactly one active Solution Pack (defining a stock feature/module list, AI tier, and quota ceilings), any number of independently priced Add-On toggles, and any number of independently priced Compliance & Security Standard toggles. Every Pack-gated route and every quota check must be enforced at the server/API boundary, independently of and in addition to RBAC — client-side navigation gating is a UX convenience only, never the security boundary.

3.4 AI Governance Architecture

A governed AI Use Case Library is the single designated integration point for any live LLM call. Every use case is versioned, tiered Assistive or Augmented, activatable per Organization and overridable per Project (Inherit/On/Off), and logged to an append-only AI Usage Log capturing each suggestion's outcome (Accepted/Edited/Rejected). A Real LLM Provider Connection, where offered, is configured and stored client-side only, is forwarded once per generation request, and is never persisted server-side, logged, or included in any export. Every AI Use Case computes a deterministic, explainable result and falls back to it automatically if a live call fails, so a misconfigured or unreachable connection never blocks a suggestion.

3.5 Retrieval-Augmented Generation (RAG) Architecture

AI-assisted suggestions, an AI Assistant's application-help mode, and any 'ask the application' chatbot are grounded by retrieval over one or more corpora — the tenant's own historical records, a Standards/Knowledge Base, the tenant's own Return on Experience (REX) entries (Section 3.10),

and/or a static feature/FAQ corpus — using a TF-IDF/cosine-similarity search or an equivalent lightweight mechanism that runs fully offline, with no network call or API key, so it is reproducible and independently verifiable. RBAC visibility rules applied to retrieval must be the same rule the rest of the application enforces, never a second, hand-maintained copy liable to drift out of sync.

3.6 Internationalization Architecture

A single, versioned string dictionary keyed by string id holds English, French, and Arabic values for every user-facing string. Language precedence resolves, on every sign-in and every Organization switch, in this order: the signed-in user's own explicit preference, then the Organization's configured default language, then English as the platform fallback. Arabic renders full right-to-left — navigation, tables, and charts mirror correctly, not merely the text direction.

3.7 Justification Governance Architecture

Every score or state change to a governed record follows a stage-then-justify pattern: the user stages the new value, writes a justification note, then saves — at which point the new value and its justification are appended together, in one atomic update, to that record's audit/change log. A platform-wide toggle determines whether the justification note is a hard requirement before Save enables, or merely offered and recorded when given.

3.8 Version Management Architecture

Any governed entity type may opt into the platform's generic Version Management service rather than implementing its own ad-hoc history mechanism: on every edit, the service duplicates the entity's current state into an immutable, timestamped version record, advances the entity's Current pointer to the new version, and never rewrites or deletes a prior version. Reverting to a historical version duplicates that version's content forward into a brand-new Current version — the same operation already defined for the AI Use Case Library (Section 4.4) and Template Libraries (Section 4.13) — so the same mechanism backs any entity type an application registers with it, rather than being re-implemented per entity.

3.9 Multi-Channel Communication Architecture

The in-app Notification Center (Section 4.9) is one of several delivery channels a Communication Dispatch service can route a message through — in-app, email, SMS, push, and webhook — behind one shared abstraction, so a new channel can be added without changing the code that raises the underlying alert or message. Every message is composed once, in the recipient's own display language (Section 3.6), and dispatched to every channel the recipient has enabled for that message category; a channel failure is logged and retried on its own schedule and never blocks delivery on another channel.

3.10 Return on Experience (REX) Architecture

Any closable entity — at minimum the application's principal record (a Project, an NCPFiche, a Change Management Project) and an individual Action — can carry a Return on Experience (REX) capture prompted at closure/completion, structured rather than left in free-form prose, and, once saved, indexed into the same retrieval corpus the Standards/Knowledge Base uses (Section 3.5) so a future AI suggestion or AI Assistant answer can be grounded in a prior REX entry exactly as it would be grounded in a Standard or the tenant's own historical records. A REX entry is itself a versioned entity

under the generic Version Management service (Section 3.8), so an edited REX entry keeps its full history like any other governed record.

3.11 Navigation Shell & Personalization Architecture

Every authenticated screen renders inside one shared navigation shell: a primary menu, a header bar carrying the Organization/Project context switcher, language switcher, notification indicator, AI Assistant entry point and user menu, and the content area. The menu is generated from a single navigation model in which each item declares its module group, route, icon, and the view permission and Solution Pack entitlement it requires — the same codes the server enforces on the route — so an item a user cannot open is never shown, and a new module appears in navigation by being registered once.

Each user chooses where the menu is docked (left, right, top, or bottom) and whether it is pinned open or slides away. Unpinned, the menu collapses to an edge handle and slides over the content when pointed at, focused, or toggled. These choices, the user's favorite items, and collapsed menu groups are stored server-side as user preferences, so they follow the user to any browser or device. Left and right are logical sides: under Arabic RTL they mirror. Below a mobile breakpoint the shell switches to a drawer, whatever the stored dock position.

3.12 Internal Benchmarking Architecture

Internal benchmarking computes a catalog of comparable metrics — each defined once with its unit, its direction of good (higher or lower is better), and its computation from the platform's own records, including the audit trail where history matters — and aggregates them by segment. Two scopes exist. Within an Organization, segments are dimensions of the Organization's own data (for example record or project type, process track, or owning department), and authorized users keep full drill-down to the underlying records. Across a Group, each participating Organization contributes only per-metric aggregates; no record, name, or identifier leaves its tenant.

Every segment or Organization below the minimum sample size is flagged as not comparable and excluded from ranks and medians. An Organization Administrator can withdraw the Organization from Group sharing at any time, after which its figures are no longer shown to the rest of the Group. Comparison with organizations outside the Group, or with external or third-party data, is out of scope.

3.13 Platform Operations Architecture

Operational robustness belongs to the platform layer, not to each deployment. On every start the server applies additive migrations — creating missing tables and columns and adding new permission codes with their default role grants — without dropping data or overwriting an administrator's edits, so an existing installation upgrades in place. Startup guards report an unsupported runtime, a port already in use, or an empty database, each as a plain-language message naming the command that fixes it, rather than a stack trace.

The persistence layer takes consistent online backups on demand and on a daily schedule, keeps them for a configurable retention period, and lists them to the platform administrator. A short-lived, per-user read cache, invalidated by any write in the same Organization, keeps read latency within target under concurrent load without ever serving one user's or one tenant's data to another.

3.14 Vertical & SME Layer Architecture

Requirements are organized in five layers, each built on the one below: Layer 1 Platform (tenancy, security, operations), Layer 2 Core (the common requirements of this standard), Layer 3 Vertical (sector content), Layer 4 SME Track (right-sized rigor for small and medium enterprises) and Layer 5 Packaging & Pricing. A vertical and an SME track are configuration data interpreted by the platform: adding, versioning, activating, rolling back or retiring them never requires a change to platform code or downtime.

Activation is additive and ordered. An Organization activates zero or more verticals; each activates its macro processes (three to five) and E2E processes (two to four). When an SME selects a vertical, the vertical processes are activated first and the SME track processes second. A project then takes a mode (Full or SME), a track recommended by its complexity score, and a template from the catalog; the template brings the phases, the Gate Library supplies the gates of each phase, and each gate brings the checklists associated with it from the Checklist Library; a checklist can also be attached to a phase without any gate.

3.15 Standards & Requirements Architecture

Standards and requirements management is built on three catalogs with one rule: a requirement exists once. The Standards Catalog holds each standard and its structure (part > chapter > clause > sub-clause, to any depth). The Requirements Catalog holds every requirement as a typed record; a standard requirement is a catalog requirement placed at a leaf of its standard's structure, and an internal, regulatory, contractual or product requirement is the same kind of record, with or without a standard. The Applicability register links standards, clauses or single requirements to an Organization and/or a project and carries the compliance assessment.

Applicability is inherited downward: what applies to an Organization applies to all its projects, and a project adds its own standards or requirements and may tailor inherited ones only with a justification; a mandatory Organization requirement cannot be set to Not applicable in a project without an approved waiver. Requirements are linked to each other (derives from, satisfies, equivalent to, conflicts with) so that one piece of evidence can prove several standards, and they are linked to the objects that implement or verify them: Controls, checklist items, gate criteria, processes, tasks and attachments. The Compliance & Security Standards modules of Section 4.3 and the vertical standards of Section 4.27 are entries of the same Standards Catalog, and the catalogs are indexed by the retrieval engine of Section 3.5.

Two ownership levels keep content governed and private: platform entries (external standards, reference frameworks) are published once and shared read-only with every Organization; Organization entries (internal frameworks, customer requirements) stay private to their tenant and may be shared read-only with the other Organizations of the same Group. Where a standard's text is copyrighted, the catalog stores the clause reference, title and a paraphrase, and the full text only under a licence recorded with the standard.

4. Common Functional Requirements

Each requirement below is common to journi and NCP Solver, generalized to platform vocabulary and stated in “the system shall” form for verifiability. Where the two source SRS documents diverge on a requirement's implementation depth, the stronger (more complete) formulation is taken as the standard; Appendix A records where a given application's current build falls short of it.

Requirements added in revision 1.1 are generalized from the CortexPLM reference build; they carry new IDs following the last 1.0 ID of their section, are listed in Appendix B, and their conformance status is recorded in Appendix A.

Requirements added in revision 1.3 in Sections 4.25 to 4.32 restate the Application-Agnostic Requirements for Verticals and SME source in platform vocabulary; duplicates in the source (for example the vertical approval workflow and audit trail stated twice) are merged, and Appendix E gives the mapping. Sections 4.33 to 4.35 come from the Dynamic Apps product direction.

4.1 Tenant & Organizational Hierarchy

- **FR-DA-TEN-01:** The system shall support full Create, Read, Update, and Delete on Group, Organization, and Project records, gated by a hierarchy-management capability.
- **FR-DA-TEN-02:** The system shall allow an Organization to optionally belong to a Group; an Organization not assigned to a Group shall operate as fully independent.
- **FR-DA-TEN-03:** Deleting a Group shall not delete its member Organizations; deleting an Organization shall cascade-delete its Projects and directly-scoped user accounts and AI activation/override state.
- **FR-DA-TEN-04:** The system shall support full Create, Read, Update, and Delete on Organizational Breakdown Structure (OBS) entries, each typed or free-text, optionally parented to another OBS entry, forming a per-Organization or per-Project tree.
- **FR-DA-TEN-05:** The system shall prevent deleting an OBS node from breaking referential integrity, either by cascading or by re-parenting/reassigning dependent records.
- **FR-DA-TEN-06:** The system shall display, for every OBS node, a live count of linked item types, so accountability and activity concentration are visible at a glance.
- **FR-DA-TEN-07:** The system shall enforce the active Solution Pack's maximum-Projects and maximum-OBS-node quotas at creation time, server-side, rejecting an over-quota request with HTTP 409.
- **FR-DA-TEN-08:** The system shall ensure that no Organization can enumerate, query, or infer the existence of another Organization through any API response, error message, or shared identifier sequence.
- **FR-DA-TEN-09:** The system shall display, on every Organization list and detail screen, whether the Organization belongs to a Group (Yes/No) and, if so, which Group; an Organization with no Group shall be labeled Independent.
- **FR-DA-TEN-10:** The system shall reject, server-side, an OBS parent assignment that crosses trees — an Organization-level node under a Project-level node, or a node under another Project's tree — and an assignment that makes a node its own ancestor.
- **FR-DA-TEN-11:** The system shall identify records by values that reveal nothing about other tenants' activity — opaque identifiers (e.g. UUIDs) or per-tenant sequences — and shall answer a request for another tenant's record exactly as for a record that does not exist (HTTP 404), in support of FR-DA-TEN-08.

- **FR-DA-TEN-12:** The system shall present a tenancy tree, Group (Yes/No) > Organization > principal records, and the same content as a table (Group Yes/No, Group, Organization, sector, records), limited to the Organizations the viewer may see.
- **FR-DA-TEN-13:** The system shall let an authorized user choose the Organization a new principal record belongs to before creating it, showing whether that Organization belongs to a Group; a platform administrator may choose any Organization, other users only their own.
- **FR-DA-TEN-14:** The system shall provision every new Organization on creation with the OBS skeleton, the standard governance catalog, the AI Use Case Library (FR-DA-AI-02), the checklist and record templates, and, when requested, a starting team of one account per standard role under the Organization's e-mail domain with an initial password; a domain already used by another Organization shall be refused.
- **FR-DA-TEN-15:** The system shall let a dedicated permission grant read-only visibility of the principal records of the other Organizations of the viewer's own Group; records of Organizations outside the Group shall never be visible, and records of another Organization shall never be editable from the viewer's Organization.
- **FR-DA-TEN-16:** The system shall let an OBS node list the people placed in it, each with a standard role and the role played in that node (for example "Engineering lead" in a project team), and shall let an administrator add and remove them.

4.2 Identity & Role-Based Access Control

- **FR-DA-RBAC-01:** The system shall support full Create, Read, Update, and Delete on user accounts, including name, email/username, one or more assigned roles, and language preference.
- **FR-DA-RBAC-02:** The system shall provide a Permission Matrix screen displaying every permission/capability code against every role, editable by users holding a dedicated permission-management capability, with edits taking effect immediately and platform-wide without a code change.
- **FR-DA-RBAC-03:** The system shall enforce every API endpoint, server-side, against the authenticated user's resolved permission set (the union across all assigned roles); a request lacking the required permission shall be rejected with HTTP 403.
- **FR-DA-RBAC-04:** The system shall enforce that the Owner and the Evaluator of any two-party verification step are never the same user, both in the UI and at the server.
- **FR-DA-RBAC-05:** The system shall provide a minimal user directory (id, name, email) accessible to any authenticated user for assignment purposes, independent of the full user-management permission.
- **FR-DA-RBAC-06:** The system shall issue a signed, time-boxed authentication token on successful login, required on every subsequent API call, and shall never expose a stored password hash or credential via any API response.
- **FR-DA-RBAC-07:** The system shall expose a single platform-wide justification-required toggle (Section 3.7), defaulting to on, gated by a hierarchy/governance-management capability.
- **FR-DA-RBAC-08:** The system shall, when a user opens a screen or deep link that their roles or the Organization's configuration do not allow, show an in-app access notice explaining why and whom to ask, never a blank page or a raw error.
- **FR-DA-RBAC-09:** The system shall provide a dedicated permission for viewing the content of individual evaluation or verification verdicts; other users, except the parties to that verification, shall see that an evaluation exists but not its content.
- **FR-DA-RBAC-10:** The system shall, at startup, add any permission code introduced by a new release to the catalog together with its default role grants, without overwriting grants an administrator has changed (Section 3.13).

4.3 Configuration Management (Solution Packs, Add-Ons, Compliance Standards)

- **FR-DA-CFG-01:** The system shall record, per Organization, exactly one Solution Pack as the single source of truth for that Organization's feature and quota entitlements, defined centrally in one configuration module and consumed by every enforcing route, so entitlements are never duplicated or able to drift between routes.
- **FR-DA-CFG-02:** The system shall define, for each Solution Pack, at minimum: which modules/features are enabled, the AI tier available (Assistive-only vs. Assistive+Augmented), and numeric ceilings on Projects, OBS nodes, and AI Use Cases.
- **FR-DA-CFG-03:** The system shall record a set of priced Add-Ons, each an independent toggle that grants its associated feature to a Pack that does not already include it standard.
- **FR-DA-CFG-04:** The system shall record a set of priced Compliance & Security Standards modules (at minimum GDPR, ISO/IEC 27001, and SOC 2), each an independent toggle available regardless of Solution Pack.
- **FR-DA-CFG-05:** The system shall, on activating a Compliance & Security Standard, idempotently seed a starting Controls scaffold tagged to that framework, without duplicating it on repeated activation, and shall not delete any Control tagged to a Standard that is later deactivated.
- **FR-DA-CFG-06:** The system shall enforce every Pack-gated feature and every Add-On/Compliance-gated feature server-side via dedicated middleware on the corresponding API route, independently of and in addition to RBAC permission checks; a request for a feature the active configuration does not include shall be rejected with HTTP 403, naming the missing feature.
- **FR-DA-CFG-07:** The system shall expose the Organization's effective configuration and the static Pack/Add-On/Compliance catalog to every authenticated user regardless of role, so client-side navigation can correctly show or hide gated menu items; this read is a UX convenience, not the enforcement boundary.
- **FR-DA-CFG-08:** The system shall expose current usage against every quota dimension, computed live from current data, and shall visually flag a quota that meets or exceeds its allotment.
- **FR-DA-CFG-09:** The system shall never represent, in any UI text, Help content, FAQ answer, or generated document, that activating a Compliance & Security Standard constitutes a certification, an external audit, or a legal attestation of compliance.
- **FR-DA-CFG-10:** The system shall allow an authorized Organization administrator to register, configure, enable, and disable a connection to an external application through the External Integration Registry (Section 3.3), independently of any other registered integration.
- **FR-DA-CFG-11:** The system shall store every external integration's credentials using the same client-side-only or encrypted-server-side pattern already required for the Real LLM Provider Connection (NFR-DA-SEC-07), never in plaintext, and never logged.
- **FR-DA-CFG-12:** The system shall verify an external integration's connection health on demand and shall record the outcome (success, authentication failure, unreachable) with a timestamp, without exposing provider-side error detail that could aid an unrelated attacker.
- **FR-DA-CFG-13:** The system shall allow an authorized administrator to map fields between an external application's records and the platform's own entities, and shall log every inbound or outbound integration transaction (record, timestamp, direction, and result) for audit (Section 4.11).
- **FR-DA-CFG-14:** The system shall accept inbound events from an external application only through the Multi-Channel Communication webhook channel (Section 4.17) or an equivalent authenticated endpoint, never an unauthenticated route, and shall scope every inbound event to the Organization that registered the originating integration.
- **FR-DA-CFG-15:** The system shall record every Compliance & Security Standards module of FR-DA-CFG-04 as an entry of the Standards Catalog (Section 4.36), so that activating the module also applies the

standard to the Organization (Section 4.38) and seeds its Controls from the standard's requirements (FR-DA-CFG-05).

4.4 AI Use Case Library & Governance

- **FR-DA-AI-01:** The system shall support full Create, Read, Update, Delete, and version-revert on an AI Use Case catalog, each entry capturing an id, name, tier (Assistive/Augmented), owning module, trigger, expected output, a human-checkpoint description, and versioned prompt-template content, gated by a dedicated AI-use-case-management capability.
- **FR-DA-AI-02:** The system shall seed every Organization with the same library of AI Use Cases on creation, and shall allow an Organization to additionally author custom AI Use Cases up to its active Solution Pack's custom-use-case quota.
- **FR-DA-AI-03:** The system shall create Version 1 of an AI Use Case automatically on creation, mark it Default/Current, and shall retain the full version history indefinitely; reverting to a historical version shall duplicate its content into a brand-new Current version, never rewrite or delete history.
- **FR-DA-AI-04:** The system shall allow an Organization Administrator to activate or deactivate any AI Use Case for the whole Organization, and shall allow an authorized Project-level user to set a tri-state (Inherit/On/Off) override per Project; effective activation shall resolve the Project override when set, else the Organization's activation state.
- **FR-DA-AI-05:** The system shall block, server-side, any new AI suggestion from a deactivated AI Use Case (by its effective state) immediately on deactivation, while never retroactively hiding or removing content a human already reviewed and approved before deactivation.
- **FR-DA-AI-06:** The system shall visually label every AI-generated output, wherever shown, as AI-generated and requiring human review, alongside its tier and its generation source (deterministic built-in engine or a configured live model).
- **FR-DA-AI-07:** The system shall write an append-only AI Usage Log entry for every AI suggestion's outcome (Accepted, Edited, or Rejected), capturing the use case, Organization, Project, the record it applied to, the acting user, and a timestamp; the log shall expose no edit or delete route.
- **FR-DA-AI-08:** The system shall allow a user to configure a Real LLM Provider Connection (provider, model, endpoint, API key) that is persisted client-side only, is forwarded once per generation request, is never returned in any API response, never written to any log, never cached or persisted server-side beyond that single request, and is never included in any data export.
- **FR-DA-AI-09:** The system shall, on any failure of a live LLM call, fall back automatically to the AI Use Case's deterministic built-in generator, so a misconfigured or unreachable connection never blocks a suggestion from being produced.
- **FR-DA-AI-10:** The system shall ensure an AI Use Case's activation, tier, or generation never grants a user visibility into data beyond what their RBAC role and permissions already allow.
- **FR-DA-AI-11:** The system shall, before generating any suggestion, ground it in retrieved reference content (Section 4.5), instruct the underlying model to stay within the application's own vocabulary, and state no number, score, or name not present in the retrieved references or the record's own data; the retrieved references shall be disclosed to the user alongside the output.
- **FR-DA-AI-12:** The system shall show every active AI Use Case, labelled with its tier (Assistive AI or Augmented AI) by a distinct icon and wording, on the processes, macro processes, tasks and steps it is linked to, and shall let an authorized user run it there and accept, modify or reject the suggestion (FR-DA-AI-07).
- **FR-DA-AI-13:** The system shall let an AI Use Case be linked to a step, a process, a task or an end-to-end process, and shall list the tasks of the process design with the AI Use Cases that support them, filterable by tier.

- **FR-DA-AI-14:** The system shall let an authorized user delete a custom AI Use Case, keeping its usage-log entries; seeded catalog entries shall be deactivated instead of deleted.
- **FR-DA-AI-15:** The system shall offer, wherever the live model is configured (personal settings and administration), a provider list and a model list with a "custom model" choice and an optional custom endpoint, and a connection test that reports only the outcome category (works, authentication failure, model not found, rate limited, unreachable).

4.5 Standards / Knowledge Base & Retrieval-Augmented Generation

- **FR-DA-KB-01:** The system shall maintain a Standards/Knowledge Base of reference content — regulatory or methodological standards, or per-process/per-stage practitioner guidance — available in every language the platform supports.
- **FR-DA-KB-02:** The system shall index this knowledge base, the tenant's own historical records, and a static application-feature/FAQ corpus using the same retrieval mechanism, so AI Use Case grounding, an AI Assistant's application-help mode, and any chatbot share one retrieval engine rather than several.
- **FR-DA-KB-03:** The system shall scope every retrieval query to the requesting user's own tenant; no cross-tenant record shall appear in a retrieval result or an AI suggestion.
- **FR-DA-KB-04:** The system shall gracefully degrade when a live LLM call is unavailable or fails: the retrieval result (the matched reference content, the computed fact) shall still be returned, rather than the request failing outright.
- **FR-DA-KB-05:** The system shall index the Standards Catalog and the Requirements Catalog with the knowledge base, so that AI grounding and search return the clause and requirement that apply, only from standards the user may see.

4.6 Process/Stage Governance (Alerts, Business Rules, Controls, Risks, RACSI, Custom KPIs)

- **FR-DA-GOV-01:** The system shall support full Create, Read, Update, and Delete on Business Rules, each with a condition, an action, a severity, and an owner, gated by a governance-management capability and, where the platform separates GRC from the base module set, by the active Solution Pack's GRC entitlement.
- **FR-DA-GOV-02:** The system shall support full Create, Read, Update, and Delete on Controls, each classified under one of the five COSO Internal Control — Integrated Framework components, with a control type, testing frequency, owner, and effectiveness rating.
- **FR-DA-GOV-03:** The system shall allow a Control to be optionally tagged with the Compliance & Security Standard it was seeded under or otherwise supports, without that tag itself constituting a certification (Section 4.3, FR-DA-CFG-09).
- **FR-DA-GOV-04:** The system shall support full Create, Read, Update, and Delete on Risk/Opportunity items, each with a type, a likelihood (1–5) and impact (1–5) rating, an automatically computed score, an owner, and a status, and shall render a 5×5 likelihood × impact heat-map.
- **FR-DA-GOV-05:** The system shall support full Create, Read, Update, and Delete on Custom KPIs, additive to the platform's built-in KPI set, each with a plain-language formula description, a target, and an owner, available regardless of Solution Pack.
- **FR-DA-GOV-06:** The system shall support a RACSI accountability matrix: full Create, Read, Update, and Delete on RACSI Activities, each optionally linked to a process step or to a specific Business Rule, Control, or Risk/Opportunity record, and on RACSI Assignments, each assigning exactly one of the five RACSI roles to a named person or a role.

- **FR-DA-GOV-07:** The system shall enforce that a RACSI Activity has at most one Accountable assignment at any time, both in the UI and at the data-persistence layer.
- **FR-DA-GOV-08:** The system shall allow a Business Rule, a Control, a Risk/Opportunity, and a RACSI Activity to each be optionally tagged with the owning OBS node and with the process/stage they are most relevant to, and shall surface the items tagged to a given process/stage on that process/stage's own governance view.
- **FR-DA-GOV-09:** The system shall let a Business Rule, a Control or a Risk/Opportunity reference one or more catalog requirements (Section 4.37), refining the standard-level tag of FR-DA-GOV-03 down to the requirement.

4.7 BPMN 2.0 Process Modeling

- **FR-DA-BPMN-01:** The system shall support full Create, Read, Update, and Delete on BPMN Diagrams, each with a title, description, and a BPMN 2.0-compliant XML payload, rendered/edited with a standards-compliant BPMN 2.0 modeler/viewer component, not a static image.
- **FR-DA-BPMN-02:** The system shall grant read-only pan/zoom viewing of a BPMN Diagram to any user holding view access, and shall grant the full editing palette and a Save action only to users holding edit access whose Organization's active configuration includes full BPMN editing.
- **FR-DA-BPMN-03:** The system shall render the BPMN editor's tool palette in a dedicated panel beside the diagram canvas, structurally separate from the canvas, rather than floating over it.
- **FR-DA-BPMN-04:** The system shall provide Export (download the current diagram as a schema-valid .bpmn/XML file) and Import (load a .bpmn/XML file, replacing the current diagram) actions, so a diagram is portable to and from any other BPMN 2.0-compliant tool.
- **FR-DA-BPMN-05:** The system shall be seeded, per Organization, with at least one BPMN Diagram modeling the application's own core process end-to-end.
- **FR-DA-BPMN-06:** The system shall offer a shapes palette section grouped by category (tools, events, activities, gateways, data, participants, artifacts), usable by click or drag, and shown as a legend of the main shapes when the diagram is view-only.
- **FR-DA-BPMN-07:** The system shall let the user open a diagram in a full-screen workspace with the palette, zoom controls (slider, zoom in and out, fit), a palette that slides away, and a single action or the Escape key to return to the page.
- **FR-DA-BPMN-08:** The system shall state on the diagram screen whether editing a diagram changes the execution of records; where diagrams only document the process, saving one shall not alter running or future records.

4.8 Reports & Data Export

- **FR-DA-REP-01:** The system shall compute the platform's standard KPIs in real time from current data, scoped to the requesting user's tenant.
- **FR-DA-REP-02:** The system shall provide a set of standard reports, each exportable on demand as a downloadable PDF, Excel (.xlsx), and Word (.docx) file, generated server-side and styled to the suite's common visual identity.
- **FR-DA-REP-03:** The system shall name every exported report file with the report key and the export date, and shall serve it with the correct MIME type and a Content-Disposition attachment header.
- **FR-DA-REP-04:** The system shall offer a client-side CSV export on its highest-traffic tables, reflecting exactly the rows and columns currently on screen and respecting the same RBAC and scope filtering as the table itself, written with a UTF-8 byte-order mark so accented and Arabic content opens correctly in common spreadsheet tools.

4.9 Notification Center (Alerts)

- **FR-DA-ALT-01:** The system shall define a shared Alert Catalog naming every alert type once, with a human-readable name and a primary process/stage tag, consumed identically by seed data and by live computation, so alert semantics cannot drift between demo data and live behavior.
- **FR-DA-ALT-02:** The system shall surface a persistent, dismissible per-tenant or per-project alert inbox, with a live unread-alert indicator in the primary navigation.
- **FR-DA-ALT-03:** The system shall avoid raising a duplicate alert of the same type against the same triggering entity within the same period.
- **FR-DA-ALT-04:** The system shall allow a user to mark an individual alert as read, and, where dismissal is supported, shall persist the dismissal separately from the alert's own (recomputed) state.
- **FR-DA-ALT-05:** The system shall allow an authorized user to independently enable or disable each alert type, displaying its human-readable name alongside its machine code in both the alert inbox and any governance-settings configuration list.

4.10 AI Assistant / Chatbot

- **FR-DA-AST-01:** The system shall provide a data-query mode that answers natural-language questions about the requesting user's own organization by matching the question to a fixed set of data-query intents.
- **FR-DA-AST-02:** The system shall associate every data-query intent with the single RBAC permission required to answer it, and shall evaluate that permission after matching the intent but strictly before running any query; when the permission is missing, the system shall return an explicit, human-readable refusal naming the missing permission rather than executing the query.
- **FR-DA-AST-03:** The system shall scope every data-query answer to the requesting user's own tenant, consistent with the platform's multi-tenancy requirement (Section 4.1).
- **FR-DA-AST-04:** The system shall provide an application-help mode answering “Can it...?” / “How do I...?” questions by ranking a static FAQ/feature corpus with the same retrieval engine used elsewhere in the platform (Section 4.5).
- **FR-DA-AST-05:** The system shall render the AI Assistant both as a dedicated screen and as a floating widget accessible from every authenticated screen, without navigating away from the user's current work.
- **FR-DA-AST-06:** The system shall gate the AI Assistant behind a single view permission and the active Solution Pack's AI Assistant entitlement, independent of and in addition to any module-specific permission a given data-query intent may separately require.

4.11 Audit Trail & Justification Governance

- **FR-DA-AUD-01:** The system shall write an audit or change-log entry for every state-changing operation on a governed field, capturing the actor, timestamp, entity type/id, and the before/after value.
- **FR-DA-AUD-02:** The system shall follow the stage-then-justify pattern (Section 3.7) for every score or state change to a governed record: staging a new value, requiring or offering a justification note per the platform-wide toggle, and appending both atomically to the record's audit/change log in one update.
- **FR-DA-AUD-03:** The system shall make the audit/change log viewable, read-only, to users holding an audit-view permission, and shall never expose an edit or delete route for it.
- **FR-DA-AUD-04:** The system shall ensure audit-log writes never block or fail the primary operation they accompany.
- **FR-DA-AUD-05:** The system shall allow an authorized user to reopen a completed workflow step before the stage or approval it belongs to is decided, only with a justification recorded per Section 3.7; the reopening shall return the step to its active state and be written to the audit log.

4.12 Help & In-App Guidance

- **FR-DA-HLP-01:** The system shall provide an in-app Help screen presenting topic articles covering every module of the application, accessible to every authenticated user independent of role or permission grants.
- **FR-DA-HLP-02:** The system shall render Help content in the user's currently selected display language, matching the article set one-to-one across every supported language.
- **FR-DA-HLP-03:** The system shall provide a free-text search box on the Help screen filtering topics by a case-insensitive match against the topic title or body.
- **FR-DA-HLP-04:** The system shall provide, next to every scored or classified input (for example complexity scores), a decision matrix describing each level, and shall show a legend for every code or colour vocabulary displayed in a list or matrix (for example the Part letter of a process or the status colours of a matrix).

4.13 Template Libraries

- **FR-DA-TPL-01:** The system shall support a library of pre-filled starting points (templates) for its principal record type, each capturing default field values a new record can be seeded from, while leaving every field editable before submission.
- **FR-DA-TPL-02:** The system shall support full Create, Read, Update, and Delete on templates, versioned via the same version-history mechanism as the AI Use Case Library (Section 4.4), gated by a dedicated template-management capability.
- **FR-DA-TPL-03:** The system shall make templates available regardless of Solution Pack, consistent with the always-standard treatment of Custom KPIs and standard Reports.
- **FR-DA-TPL-04:** Deleting a template shall not affect any record already created from it; a template is a starting point copied at use time, not a live-linked reference.
- **FR-DA-TPL-05:** The system shall manage project templates through the Project Template Catalog of Section 4.33; the rules of FR-DA-TPL-01 to 04 apply to it.

4.14 Multi-Language & Localization

- **FR-DA-I18N-01:** The system shall provide complete UI translation in English, French, and Arabic for all navigation, labels, form fields, statuses, and messages, with no hardcoded UI string outside the shared translation dictionary.
- **FR-DA-I18N-02:** The system shall allow a user to change their display language at any time without reloading or losing unsaved work.
- **FR-DA-I18N-03:** The system shall persist a user's language preference and apply it automatically on subsequent logins, resolved per the precedence order in Section 3.6.
- **FR-DA-I18N-04:** The system shall render the entire application right-to-left, including navigation, forms, tables, and charts, when Arabic is selected.
- **FR-DA-I18N-05:** The system shall allow an Organization to configure a tenant-level default language.
- **FR-DA-I18N-06:** The system shall allow adding a new supported language by configuration — populating the shared translation dictionary (Section 3.6) — alone, without a code change or a redeployment of application logic.
- **FR-DA-I18N-07:** The system shall return every server-generated, user-facing message — validation and error messages, notification text, AI Assistant answers, and export labels — in the requesting user's display language, resolved from the same translation dictionary as the UI; a machine code may accompany the translated text but shall not replace it.

- **FR-DA-I18N-08:** The system shall seed its reference and Knowledge Base content in every supported language and shall let the user filter Knowledge Base articles by language, defaulting to the user's display language.

4.15 Work Breakdown Structure & Gantt Scheduling

Any grouping of Actions — regardless of the process, stage, or governed record they originate from — can be organized into a Work Breakdown Structure (WBS) and scheduled as a Gantt chart, so planning is available for any action grouping the user assembles, not only a single predefined project type.

- **FR-DA-WBS-01:** The system shall allow any set of Actions the user selects to be grouped into a Work Breakdown Structure (WBS), organized as a tree of WBS nodes, each node referencing one or more Action IDs as its leaf-level work items.
- **FR-DA-WBS-02:** The system shall allow a WBS node or a referenced Action to carry a planned start date, planned end date or duration, percent-complete, and zero or more predecessor dependencies on other WBS nodes or Actions within the same WBS.
- **FR-DA-WBS-03:** The system shall render a WBS as a Gantt chart, plotting every node and its referenced Actions on a shared timeline, drawing dependency lines between predecessor and successor items, and visually distinguishing planned, in-progress, completed, and overdue items.
- **FR-DA-WBS-04:** The system shall recompute a WBS node's own planned dates and percent-complete from its child nodes/Actions when the node itself is not directly scheduled, so a summary bar always reflects the current state of its children without manual reconciliation.
- **FR-DA-WBS-05:** The system shall support full Create, Read, Update, and Delete on WBS structures, and shall allow the same Action to be referenced from more than one WBS without duplicating the underlying Action record.
- **FR-DA-WBS-06:** The system shall export a WBS/Gantt view to at least one portable format (e.g. PDF or an image), consistent with the platform's standard Reports & Data Export conventions (Section 4.8).

4.16 Version Management

- **FR-DA-VER-01:** The system shall provide a single, generic Version Management service (Section 3.8) that any governed entity type can register with, rather than each entity type implementing its own history mechanism.
- **FR-DA-VER-02:** The system shall create an initial version automatically when a versioned entity is created, mark it Current, and create a new version — never overwrite the prior one — on every subsequent edit.
- **FR-DA-VER-03:** The system shall retain every version of a versioned entity indefinitely, with no hard-delete route for a historical version, and shall allow an authorized user to view the full version history and compare any two versions field-by-field.
- **FR-DA-VER-04:** The system shall allow an authorized user to revert a versioned entity to any historical version; reverting shall duplicate that version's content into a brand-new Current version, never rewrite or delete the entity's history, consistent with the pattern already defined for the AI Use Case Library (FR-DA-AI-03) and Template Libraries (FR-DA-TPL-02).
- **FR-DA-VER-05:** The system shall record, on every version, the acting user, a timestamp, and — where the platform-wide justification toggle (Section 3.7) requires it — the justification note that accompanied the change.
- **FR-DA-VER-06:** The system shall provide one shared version-comparison view, available from the history of every versioned entity, that shows two selected versions side by side and highlights the fields that differ, in support of FR-DA-VER-03.

4.17 Multi-Channel Communication

- **FR-DA-COMM-01:** The system shall support delivering a notification or alert (Section 4.9) through one or more channels — at minimum in-app, email, and webhook, with SMS and push as optional channels — behind a single shared dispatch interface (Section 3.9).
- **FR-DA-COMM-02:** The system shall allow a user to independently enable or disable each available channel for each notification category, defaulting every category to the in-app channel at minimum.
- **FR-DA-COMM-03:** The system shall compose a message once, per notification event, and render it in the recipient's own display language (Section 3.6) before dispatch to any channel.
- **FR-DA-COMM-04:** The system shall record a delivery-status record per message per channel (queued, sent, delivered, failed), and shall retry a failed delivery on that channel's own schedule without blocking or duplicating delivery on any other channel.
- **FR-DA-COMM-05:** The system shall never send a tenant's data through an external channel provider (email, SMS, push) beyond the minimum content required for that notification, and shall scope every dispatch to the recipient's own tenant.

4.18 Return on Experience (REX)

A Return on Experience (REX) entry captures what a closed Project (or equivalent principal record) or a closed Action actually taught the organization, so the lesson is queryable and reusable rather than lost in an individual's memory or a narrative report.

- **FR-DA-REX-01:** The system shall prompt for a REX entry when the application's principal record reaches a terminal/closed status, and when an individual Action is marked complete, without blocking the closure itself on the REX entry being filled in.
- **FR-DA-REX-02:** The system shall structure every REX entry with, at minimum: what went well, what did not, a root-cause note, a recommendation/improvement action, a category/tag, and an effectiveness or satisfaction rating on a fixed scale.
- **FR-DA-REX-03:** The system shall support full Create, Read, Update, and Delete on REX entries, each optionally tagged to the owning OBS node and to the process/stage it is most relevant to, consistent with how other governance artifacts are tagged (Section 4.6, FR-DA-GOV-08).
- **FR-DA-REX-04:** The system shall index every saved REX entry into the same retrieval corpus used for AI Use Case grounding, the AI Assistant, and application-help search (Section 4.5), scoped to the requesting user's own tenant, so a relevant prior REX entry can surface in a future suggestion or answer.
- **FR-DA-REX-05:** The system shall provide a REX Register view aggregating entries by category/tag and by rating, so a recurring root cause or a declining rating trend is visible without reading every entry individually.
- **FR-DA-REX-06:** The system shall version every REX entry via the generic Version Management service (Section 3.8/4.16), retaining its full edit history and supporting revert to any prior version.
- **FR-DA-REX-07:** The system shall gate REX entry management behind a dedicated capability, and shall scope every REX read and every retrieval-grounded reference to it to the requesting user's own tenant, consistent with the platform's multi-tenancy requirement (Section 4.1).
- **FR-DA-REX-08:** The system shall present the REX prompt to the user who performed the closing action, immediately after the closure succeeds, with a choice to fill it in now or skip; skipping shall leave the closure intact and the entry available to add later from the REX Register.

4.19 Navigation Shell & Personalization

The navigation shell (Section 3.11) behaves the same way in every Dynamic App, so a user who has learned one application can find their way around any other.

- **FR-DA-NAV-01:** The system shall generate the primary menu from a single navigation model in which each item declares its module group, label, icon, route, and the view permission and Solution Pack entitlement it requires, and shall show only the items the current user can open.
- **FR-DA-NAV-02:** The system shall group menu items by module, let the user collapse and expand each group, and mark the current screen as the active item.
- **FR-DA-NAV-03:** The system shall let each user pin the menu open or unpin it. Unpinned, the menu shall collapse to an edge handle, slide over the content when the user points at the edge, focuses the handle, or uses the menu button, and slide away again when the pointer leaves or an item is chosen.
- **FR-DA-NAV-04:** The system shall let each user dock the menu on the left, right, top, or bottom edge of the screen, laying it out vertically on the left and right and horizontally, with drop-down groups, on the top and bottom.
- **FR-DA-NAV-05:** The system shall let each user mark menu items as favorites, shown in a Favorites group at the start of the menu.
- **FR-DA-NAV-06:** The system shall store the dock position, pinned state, collapsed groups, and favorites as server-side user preferences, applied on every sign-in and on any device, and shall offer them both from the menu itself and from the user's Settings screen.
- **FR-DA-NAV-07:** The system shall treat left and right as logical start and end under right-to-left languages, mirroring the dock position and the slide direction when Arabic is selected.
- **FR-DA-NAV-08:** The system shall, below a mobile breakpoint, replace the docked menu with a drawer opened from a menu button and closed by Escape, by an outside tap, or by choosing an item, without changing the user's stored desktop preferences.
- **FR-DA-NAV-09:** The system shall make the shell fully keyboard-operable: a skip-to-content link as the first focusable element, menu and group buttons exposing their expanded state, Escape closing any open menu, and the active item marked as the current page for assistive technologies.
- **FR-DA-NAV-10:** The system shall keep the Organization/Project context switcher, language switcher, notification indicator, AI Assistant entry point, and user menu in the header bar of every authenticated screen.
- **FR-DA-NAV-11:** The system shall organize the menu by what users do, in a documented order (home and own work, portfolio of records, process design, governance and risk, AI and knowledge, reports, administration, personal settings), so the same capability is always found in the same place across Dynamic Apps.

4.20 Internal Benchmarking

Internal benchmarking (Section 3.12) lets an Organization learn from its own segments and from the other Organizations of its Group. External benchmarking is out of scope.

- **FR-DA-BMK-01:** The system shall define its benchmark metrics in one catalog module, each with a label, a unit, a direction of good, a plain-language explanation, and a computation from the platform's own records, consumed by every benchmarking screen.
- **FR-DA-BMK-02:** The system shall compare, within one Organization, every catalog metric across the segments of at least one dimension the application defines (for example record or project type, process track, or owning OBS node), showing each segment's value, the Organization-wide value, and the median.

- **FR-DA-BMK-03:** The system shall compare, across the Organizations of the requester's Group, per-Organization aggregates of the same metrics, and shall rank each Organization per metric according to the metric's direction of good, giving tied values the same rank.
- **FR-DA-BMK-04:** The system shall include in a Group comparison only per-metric aggregates and each contributing Organization's name and sector — never a record, record title, person's name, or identifier from another Organization.
- **FR-DA-BMK-05:** The system shall flag any segment or Organization with fewer records than a minimum sample size (default 3) as not comparable, and shall exclude it from ranks and medians.
- **FR-DA-BMK-06:** The system shall let an Organization Administrator withdraw the Organization from Group benchmarking and rejoin it; while withdrawn, its figures shall not be shown to the other Organizations of the Group. Every change shall be written to the audit log.
- **FR-DA-BMK-07:** The system shall gate internal benchmarking behind a view capability, Group comparison behind a separate capability, and the sharing setting behind a management capability, each enforced server-side.
- **FR-DA-BMK-08:** The system shall refuse any comparison with an Organization outside the requester's Group and shall not import or display external or third-party benchmark data; an Organization with no Group shall see its internal comparison only, with a notice explaining why.
- **FR-DA-BMK-09:** The system shall render each metric comparison as a chart following the common visual identity, with the requester's own Organization or the selected segment highlighted, and shall let the user filter the comparison by the application's segment dimensions.

4.21 Platform Operations

These requirements keep a Dynamic App safe to upgrade, recover, and demonstrate, whatever its business domain (Section 3.13).

- **FR-DA-OPS-01:** The system shall take a consistent backup of its data store on demand — from the command line and from the platform administrator's configuration screen — and automatically once a day, while remaining online.
- **FR-DA-OPS-02:** The system shall keep backups for a configurable retention period (default 14 days), delete older ones automatically, and list existing backups with date and size to the platform administrator only.
- **FR-DA-OPS-03:** The system shall document how to restore a backup, and a restored backup from an earlier release shall start without any manual migration step (FR-DA-OPS-04).
- **FR-DA-OPS-04:** The system shall apply additive schema and permission-catalog migrations automatically on startup, never dropping or rewriting existing data or administrator changes, so an existing installation upgrades in place.
- **FR-DA-OPS-05:** The system shall check, when installed and started, the runtime version, port availability, and whether the data store has been initialized, and shall report each problem as a plain-language message naming the corrective action rather than a stack trace.
- **FR-DA-OPS-06:** The system shall expose an unauthenticated health endpoint reporting only liveness, whether the data store is initialized, and the deployment mode — never tenant data.
- **FR-DA-OPS-07:** The system shall ship a deterministic demonstration seed, reset by one command, containing at least one Group of two or more Organizations, at least one Independent Organization, several sectors, a user for every role, and at least ten instances of each principal end-to-end process per Organization, all fictional (Section 2.7).

- **FR-DA-OPS-08:** The system shall include an administrator traceability view mapping each requirement ID of this standard to its implementation status (Met, Partial, Not met, or Deployment responsibility) and a short evidence note.
- **FR-DA-OPS-09:** The system shall seed demonstration data in every module, so that no screen of the delivered application is empty: record teams with roles, templates, attachments in several formats, custom AI Use Cases and project overrides, alert settings and alerts already read, and integration and webhook registrations.

4.22 Portfolio Overview

The portfolio overview answers "where does each record stand in each process?" for any scope the user may see.

- **FR-DA-PFO-01:** The system shall let the user choose a scope: a Group, one or more Organizations, or a set of principal records, each list containing only what the user may see (FR-DA-TEN-15).
- **FR-DA-PFO-02:** The system shall show one row per principal record and one column per process stage, with at each intersection the status of that stage for that record, drawn from one status vocabulary (for example Completed, In progress, At gate, On hold, Stopped, Not started, Not in the record's variant).
- **FR-DA-PFO-03:** The system shall show a legend of the cell statuses, a per-stage total, the number of times a stage ran when more than once, and a hover text naming record, stage and status.
- **FR-DA-PFO-04:** The system shall export the matrix to CSV (FR-DA-REP-04) and shall show records of another Organization read-only.

4.23 Checklist Library

The Checklist Library holds every checklist as an independent entry. A checklist may be related to a gate or not: when it is associated with a gate of the Gate Library (Section 4.35) it becomes part of that gate's exit criteria; when it is not, it can still be attached to a phase, a task or a project and used on its own.

- **FR-DA-CHK-01:** The system shall keep a Checklist Library with full Create, Read, Update, Delete and Duplicate on checklist templates, each with ordered items marked mandatory and evidence-required, classified by scope (universal or one vertical) and optionally by mode and track, and versioned under the generic Version Management service (Section 3.8). A checklist shall not require a gate.
- **FR-DA-CHK-02:** The system shall let a checklist template be related to zero, one or several gates; the association is made from the gate (FR-DA-GTE-02) or from the checklist, and a checklist associated with a gate shall be copied into that gate's checklist when the phase opens. A built-in reference checklist shall apply to a gate that has no associated checklist.
- **FR-DA-CHK-03:** The system shall let an authorized user add items to an open gate or phase checklist from any template or by typing them, never adding the same item twice, and save such a checklist as a new template.
- **FR-DA-CHK-04:** The system shall show for every checklist the gates it is associated with, and let the library be filtered by gate, by "not related to a gate", by vertical, mode and track; it shall seed sector checklists for new Organizations of that sector.
- **FR-DA-CHK-05:** The system shall freeze a checklist, its items and their evidence once the gate it belongs to has been submitted for decision or, for a checklist not related to a gate, once it has been signed off.
- **FR-DA-CHK-06:** The system shall let a checklist that is not related to any gate be attached directly to a phase, a task or a project, with its completion tracked and reported; such a checklist shall not block any progression.

- **FR-DA-CHK-07:** The system shall show, before a checklist is changed or deleted, the gates, templates and open phases that use it; a checklist associated with a published gate shall be retired rather than deleted, and deleting a checklist shall never alter the copies already made in projects.
- **FR-DA-CHK-08:** The system shall let a checklist item and a gate criterion reference one or more catalog requirements, so that completing the item or passing the gate records evidence for those requirements (Section 4.38).

4.24 Attachments

Actions and checklist items carry the documents that support them.

- **FR-DA-ATT-01:** The system shall let a user attach several files at once to an action or a checklist item, by file picker or drag-and-drop.
- **FR-DA-ATT-02:** The system shall accept a published list of formats — documents, spreadsheets, presentations, images, drawings and CAD, data and models, archives, audio and video, e-mails — with a size and count limit per upload, and shall refuse any other type with a message naming the refused file.
- **FR-DA-ATT-03:** The system shall show for every attachment its type, size, author and date, with download for every viewer and removal for its author or a manager, audited, and never once the stage is submitted.
- **FR-DA-ATT-04:** The system shall store attachments outside every web-served directory, scoped to the tenant, and include them in the backup instructions (NFR-DA-SEC-13, FR-DA-OPS-01).

4.25 Vertical Framework & Governance

A vertical is a first-class, governed configuration entity. Its content is data, so any Dynamic App can serve new sectors without code changes.

- **FR-DA-VRT-01:** The system shall support the definition of verticals, each with a unique identifier, a short prefix, a name, a description, its target segments, its value proposition, and its associated macro and E2E processes.
- **FR-DA-VRT-02:** The system shall support a taxonomy of verticals with parent-child and sibling relationships.
- **FR-DA-VRT-03:** The system shall let administrators and partners define new verticals without modifying the platform, and new verticals shall work alongside existing ones.
- **FR-DA-VRT-04:** The system shall manage each vertical through the lifecycle Draft, Review, Approved, Active, Deprecated and Retired, with an approval workflow (configurable approvers, escalation and notifications) for definition, activation, modification, deprecation and retirement.
- **FR-DA-VRT-05:** The system shall version-control vertical definitions under the Version Management service (Section 3.8), so that every change can be tracked, reviewed, approved and rolled back, including the rollback of an activation or configuration change to a previous version.
- **FR-DA-VRT-06:** The system shall let each tenant activate its own set of verticals without affecting other tenants.
- **FR-DA-VRT-07:** The system shall refuse to deactivate a vertical, a macro process or an E2E process while active projects depend on it, shall require approval for a deactivation, and shall preserve all data of a deactivated vertical.
- **FR-DA-VRT-08:** The system shall keep an immutable, searchable and exportable audit trail of every vertical action — definition, activation, modification, deactivation and retirement — retained per policy (Section 4.11).

- **FR-DA-VRT-09:** The system shall record a designated owner for every vertical and shall support change management for it: change request, impact analysis, approval and tracked implementation.
- **FR-DA-VRT-10:** The system shall capture and report governance metrics per vertical: adoption, compliance and performance.

4.26 Vertical Processes

Selecting a vertical activates its own processes, built on the common process model of the application.

- **FR-DA-VPR-01:** The system shall activate the macro processes of a vertical when the vertical is activated, with a configurable range of three to five per vertical.
- **FR-DA-VPR-02:** The system shall describe every vertical macro process with an identifier, name, description, goals, input suppliers, inputs, tasks, outputs, output customers, related E2E processes, related modules and applicable compliance requirements.
- **FR-DA-VPR-03:** The system shall map each vertical macro process to the regulatory and industry standards that apply to it; the mapping shall be configurable and audited.
- **FR-DA-VPR-04:** The system shall let administrators and partners add vertical macro processes without modifying the platform, versioned and reversible like verticals (FR-DA-VRT-05).
- **FR-DA-VPR-05:** The system shall support tasks within each macro process (owner, deadline, dependencies, status) and steps within each task (owner, inputs, outputs, status), and shall enforce the dependencies defined between macro processes, tasks and steps.
- **FR-DA-VPR-06:** The system shall activate the E2E processes of a vertical when the vertical is activated, with a configurable range of two to four per vertical.
- **FR-DA-VPR-07:** The system shall enforce the chain relationships between E2E processes — sequential, branching, parallel, loop-back and terminal — and shall support, per E2E process, triggers (business event, time-based, manual) and terminal states (success, failure, cancellation, loop-back).
- **FR-DA-VPR-08:** The system shall support user-facing tasks in each E2E process, each linked to the macro processes, tasks and steps it relies on.
- **FR-DA-VPR-09:** The system shall capture metrics for each vertical macro and E2E process — cycle time, completion rate, quality and compliance — and produce status, compliance and performance reports from them (Section 4.8).

4.27 Vertical Data, Compliance & Integration

Each vertical may extend the data model, carry its own compliance obligations and connect to sector systems, always without breaking the common core.

- **FR-DA-VDT-01:** The system shall let a vertical extend the common data model with its own objects, attributes, relationships, validation rules and lifecycle states, without altering or breaking the common entities (Section 6).
- **FR-DA-VDT-02:** The system shall exchange vertical data in industry-standard formats, at least JSON, XML, CSV and PDF, and, where the vertical needs it, engineering formats such as STEP, IGES and JT.
- **FR-DA-VDT-03:** The system shall apply per-vertical retention policies and support archiving and retrieval of vertical data, in line with the regulatory retention of the sector.
- **FR-DA-VCP-01:** The system shall support vertical-specific regulations and standards as configurable, audited and reportable compliance requirements, subject to FR-DA-CFG-09.

- **FR-DA-VCP-02:** The system shall monitor compliance per vertical, alert stakeholders of non-compliance through the Notification Center (Section 4.9), and produce compliance reports, audit reports and certification records.
- **FR-DA-VCP-03:** The system shall support internal and external compliance audits — planning, execution, findings and corrective actions — and vertical compliance training with content, delivery, tracking and certification.
- **FR-DA-VCP-04:** The system shall hold vertical regulations and standards in the Standards Catalog and link them to their vertical, so that activating a vertical proposes them for applicability to the Organization.
- **FR-DA-VIN-01:** The system shall connect vertical-specific external systems through standard interfaces (REST, SOAP, GraphQL, file-based, message-based), under the integration rules of FR-DA-CFG-10 to 14.
- **FR-DA-VIN-02:** The system shall secure, monitor (performance, errors, data quality) and test every vertical integration, and shall handle integration errors by retry, fallback and notification.

4.28 Vertical Configuration

Vertical parameters are configured, validated and versioned before activation.

- **FR-DA-VCF-01:** The system shall let an administrator configure, per vertical: macro and E2E process activation, checklist templates, gate criteria and data model extensions.
- **FR-DA-VCF-02:** The system shall provide a pre-built configuration template per vertical for rapid deployment.
- **FR-DA-VCF-03:** The system shall validate a vertical configuration for completeness, consistency and compliance before activation, and shall refuse an activation that fails validation, naming each failed rule.
- **FR-DA-VCF-04:** The system shall version every vertical configuration change and audit who changed what, when and why.

4.29 SME Mode & Tracks

The SME mode is a cross-sector enabler: an SME in any vertical can run it alongside that vertical.

- **FR-DA-SME-01:** The system shall define an SME by configurable employee thresholds per region (default 1–250 in the EU, up to 500 under the US SBA definition) and shall segment SMEs into Micro (1–10), Small (10–50) and Medium (50–250).
- **FR-DA-SME-02:** The system shall let an Organization activate the SME mode together with any vertical, and each tenant shall activate its own tracks and configuration without affecting other tenants.
- **FR-DA-SME-03:** The system shall support two to four configurable SME tracks, each defining its number of macro processes, E2E processes, gates, checklist items per gate, and typical duration.
- **FR-DA-SME-04:** The system shall activate three to five SME macro processes and two to six SME E2E processes when a track is selected, with the same taxonomy, task, step, chain and metric rules as vertical processes (FR-DA-VPR-02, 05, 07, 08 and 09) plus track applicability and cost metrics.
- **FR-DA-SME-05:** The system shall let administrators define new SME tracks, macro processes and configurations without modifying the platform.
- **FR-DA-SME-06:** The system shall assign a project in SME mode to the track recommended by its complexity score (Section 4.30); a manual override shall require a written justification and an approval, and shall be audited and reportable.
- **FR-DA-SME-07:** The system shall capture and report, per track, its usage, effectiveness and outcomes.

- **FR-DA-SME-08:** The system shall govern the SME framework like verticals: a designated owner, an approval workflow for track definition, activation, modification, deprecation and retirement, change management, an immutable audit trail and adoption, compliance and performance metrics.

4.30 Complexity Scoring

A complexity score gives every project the right amount of rigor, consistently and traceably.

- **FR-DA-SCO-01:** The system shall compute a complexity score for every project from weighted criteria and shall use it to recommend the mode and track.
- **FR-DA-SCO-02:** The system shall let an administrator configure the criteria — at least strategic impact, technical novelty, regulatory burden, investment size, cross-functional scope, market uncertainty and time-to-market pressure — and their weights, which shall sum to 100 %; a set of weights that does not sum to 100 % shall be refused.
- **FR-DA-SCO-03:** The system shall let each vertical add its own complexity drivers (for example regulatory burden, industry standards, sector-specific risks) with their weights, versioned, so that the score and the recommended track reflect the vertical of the project.
- **FR-DA-SCO-04:** The system shall keep an audit trail of every score — criteria, weights, values, result and track assignment — and shall report the distribution of scores, track assignments and outcomes.

4.31 SME Onboarding & Experience

SMEs need to go live quickly with little PLM experience.

- **FR-DA-ONB-01:** The system shall run an onboarding plan for an SME within a configurable period (default 30 days): tenant provisioning (FR-DA-TEN-14), track configuration, data import, validation, user training and go-live.
- **FR-DA-ONB-02:** The system shall import data from spreadsheets, CSV files, legacy systems and cloud storage, preserving data integrity and relationships, and shall validate the imported data for completeness, consistency and accuracy before go-live, listing every rejected record with its reason.
- **FR-DA-ONB-03:** The system shall provide self-service training and guided onboarding: interactive tutorials, guided workflows, sample data and contextual help (Section 4.12).
- **FR-DA-ONB-04:** The system shall offer SME users simplified navigation, a mobile-responsive or progressive web app experience, and self-service support through the knowledge base, the AI Assistant (Section 4.10) and a community space.
- **FR-DA-ONB-05:** The system shall measure onboarding: time-to-value, data quality, user adoption and satisfaction.

4.32 Packaging & Pricing

SME packs, vertical packs and their bundles extend the Solution Pack model of Section 4.3.

- **FR-DA-PKG-01:** The system shall let an administrator define three to five SME packs, each with configurable contents, target segment and price, and bundles of a vertical pack with an SME pack, each recorded as a Solution Pack variant so that entitlement stays enforced by FR-DA-CFG-06.
- **FR-DA-PKG-02:** The system shall enforce configurable pricing rules — minimum price per user per month, minimum user count per pack, volume discounts and bundle discounts — and shall show how each price was obtained.
- **FR-DA-PKG-03:** The system shall support upgrading and downgrading between packs and bundles with prorated billing and without loss of data.

- **FR-DA-PKG-04:** The system shall report revenue, discounts, adoption, upgrades and downgrades per pack and bundle, exportable to PDF, XML and CSV.

4.33 Project Template Catalog

Every Dynamic App offers a catalog of project templates. A template is a starting point copied at creation time (FR-DA-TPL-04).

- **FR-DA-PTC-01:** The system shall keep a Project Template Catalog in which every template is classified by scope — Universal (common to all sectors) or one Vertical — and by mode — Full, or SME with its track.
- **FR-DA-PTC-02:** The system shall support full Create, Read, Update, Delete and Duplicate on project templates, versioned under Section 3.8; a template shall carry its default fields, its phases, the gates (Section 4.35) and checklists (Section 4.23) attached to each phase, its OBS roles and its milestones.
- **FR-DA-PTC-03:** The system shall present the catalog as a browsable list with search and filters on scope, vertical, mode, track and status, a preview of each template's phases, gates and checklists, and its use count.
- **FR-DA-PTC-04:** The system shall manage each template through Draft, Published and Retired; only published templates shall be offered at project creation, and retiring a template shall not change any project created from it.
- **FR-DA-PTC-05:** The system shall provide universal templates for Full and SME mode on every installation, seed vertical templates when a vertical is activated, and let an authorized user save an existing project as a new template.
- **FR-DA-PTC-06:** The system shall restrict catalog changes to a dedicated permission, audit every change, and let platform templates be shared read-only with all Organizations while Organization templates stay private to their tenant.

4.34 Project Creation Modes

A project can be created in three ways; all three end in the same, fully editable project.

- **FR-DA-PCM-01:** The system shall offer three creation modes on one entry point: From the catalog, Manual, and With AI.
- **FR-DA-PCM-02:** From the catalog: the system shall propose the templates that match the Organization's verticals and the project's mode and track, and copy the chosen template's fields, phases, gates, checklists and roles into the new project.
- **FR-DA-PCM-03:** Manual: the system shall let the user enter the project and choose its vertical (or none), mode and track, and build its phases by attaching gates from the Gate Library and checklists from the Checklist Library, or by creating new ones.
- **FR-DA-PCM-04:** With AI: the system shall draft the project — fields, complexity scores, recommended mode and track, closest template, phases, gates and checklists — from a short description, as an Assistive AI suggestion that the user accepts, modifies or rejects item by item before anything is saved (FR-DA-AI-07); the draft shall work with the built-in engine and use the live model when configured.
- **FR-DA-PCM-05:** The system shall compute the complexity score (Section 4.30) in every creation mode, apply the quota checks of NFR-DA-SEC-08, and record in the audit trail the creation mode, the template and version used, and the AI suggestions accepted.

4.35 Gate Library

The Gate Library holds every gate definition. Checklists from the Checklist Library (Section 4.23) are associated with a gate; gates are then attached to any phase of any project or project template.

- **FR-DA-GTE-01:** The system shall keep a Gate Library with full Create, Read, Update, Delete and Duplicate on gate definitions, versioned: name, purpose, entry and exit criteria, approvers (roles or people), decision options Go, No-Go, Hold and Recycle, and applicability by vertical, mode and track.
- **FR-DA-GTE-02:** The system shall let an authorized user associate zero, one or several checklists of the Checklist Library with a gate, optionally per track, each marked mandatory or optional, in a chosen order; a checklist created from the gate screen shall be saved in the Checklist Library, and the same checklist may be associated with several gates.
- **FR-DA-GTE-03:** The system shall let an authorized user attach one or more gates to any phase of a project or a project template, reorder them, and detach them while the phase has not started; attaching a gate brings its associated checklists with it.
- **FR-DA-GTE-04:** The system shall configure the gates of a project from its vertical and track: the number of gates depends on the track and the criteria on the vertical, and a gate may be marked optional for a track.
- **FR-DA-GTE-05:** The system shall block the progression of a phase while a mandatory item of a mandatory checklist associated with its gate is incomplete; enforcement shall be configurable per gate and a waiver shall require justification and approval.
- **FR-DA-GTE-06:** The system shall capture every gate decision with its approvers, criteria results and comments, and keep an immutable audit trail of gate actions — creation, modification, checklist association, attachment to a phase, review, decision and waiver.
- **FR-DA-GTE-07:** Changing a gate, or a checklist associated with it, shall not alter the copies already attached to started phases; the user shall be offered to update not-yet-started phases to the new version.

4.36 Standards Catalog

The Standards Catalog holds every standard the Organization works with, internal or external, structured from chapter to requirement.

- **FR-DA-STD-01:** The system shall keep a Standards Catalog with full Create, Read, Update, Delete and Duplicate on standards, each with a code, title, category (Internal or External), kind (international or national standard, regulation, industry standard, customer standard, internal framework, policy, method), issuer, edition or version, publication and effective dates, language, owner, licence note, and applicable sectors or verticals.
- **FR-DA-STD-02:** The system shall let an authorized user build the structure of a standard as an ordered tree of parts, chapters, clauses and sub-clauses to any depth, each with a number, title and optional guidance, and place requirements of the Requirements Catalog (Section 4.37) at its leaves; nodes can be added, moved, renumbered and removed while the standard is in Draft.
- **FR-DA-STD-03:** The system shall manage each standard through Draft, Published, Superseded and Withdrawn; only published standards shall be applicable, and a new edition shall be created from the previous one, with a comparison listing added, changed and removed requirements.
- **FR-DA-STD-04:** The system shall import a standard's structure and requirements from a spreadsheet (CSV or XLSX, one row per node) or ReqIF, validating numbering and parent references and reporting every rejected row, and export any standard in the same formats and to PDF.
- **FR-DA-STD-05:** The system shall let an authorized user draft a standard's structure and requirements from an uploaded document with AI, as an Assistive suggestion reviewed node by node (FR-DA-AI-07), and shall record the source document and page of every requirement so drafted.
- **FR-DA-STD-06:** The system shall publish platform standards read-only to every Organization, keep Organization standards private to their tenant with optional read-only sharing inside the Group, restrict catalog changes to a dedicated permission, and audit every change.

- **FR-DA-STD-07:** The system shall present the catalog with search and filters (category, kind, issuer, sector, status, applied or not) and, for each standard, its structure as a collapsible tree with requirement counts per node and the Organizations and projects it applies to.

4.37 Requirements Catalog

Every requirement — from a standard, an internal framework, a regulation, a contract or a product specification — is one typed record, stored once and reused everywhere.

- **FR-DA-RQM-01:** The system shall keep a Requirements Catalog with full Create, Read, Update, Delete and Duplicate on requirements, each with a unique identifier, title, statement, type, obligation level (Mandatory "shall", Recommended "should", Optional "may"), rationale, verification method (inspection, analysis, demonstration, test, audit, document review), expected evidence, source (standard and clause, document or person), owner, tags, status and version.
- **FR-DA-RQM-02:** The system shall let an administrator manage the list of requirement types with CRUD, seeded with Standard, Internal, Regulatory, Contractual, Product, Process, Security & Privacy and Quality; a type in use shall be deactivated rather than deleted.
- **FR-DA-RQM-03:** The system shall let a requirement exist with or without a standard; a requirement placed in a standard's structure shows its clause path, and the same requirement shall not be duplicated when it is reused in another standard or project.
- **FR-DA-RQM-04:** The system shall let a requirement be linked to other requirements as derives from, refines, satisfies, equivalent to or conflicts with, and show the links both ways, so that an internal requirement can be traced to the external clauses it covers.
- **FR-DA-RQM-05:** The system shall show for every requirement where it is used: standards and clauses, Organizations and projects it applies to, and the Controls, checklist items, gate criteria, processes, tasks and attachments linked to it.
- **FR-DA-RQM-06:** The system shall manage each requirement through Draft, Approved and Retired, version every change under Section 3.8, and never alter an assessment already recorded against an earlier version.
- **FR-DA-RQM-07:** The system shall present the catalog with search and filters (type, obligation, source standard, status, owner, tag, applied or not), bulk edit of type, owner and tags, and CSV, XLSX and ReqIF import and export.
- **FR-DA-RQM-08:** The system shall suggest with AI, for review, duplicate or equivalent requirements and links between requirements of different standards; no link shall be saved without a person accepting it.

4.38 Applicability & Compliance

Standards and requirements apply to an Organization, to a project, or to both; compliance is assessed where they apply.

- **FR-DA-APL-01:** The system shall let an authorized user apply a whole standard, a clause with all requirements below it, or single requirements to an Organization, to one or more projects, or to both, and shall show on the Organization and on each project the standards and requirements that apply.
- **FR-DA-APL-02:** The system shall make requirements applied to an Organization apply to all its projects; a project may add standards and requirements, and may mark an inherited requirement Tailored or Not applicable only with a justification, and, for a mandatory one, an approved waiver.
- **FR-DA-APL-03:** The system shall let a project template (Section 4.33) and a vertical (Section 4.25) carry default standards and requirements, applied to a project created from them.

- **FR-DA-APL-04:** The system shall record, for every applicable requirement in an Organization or a project, its compliance status, owner, due date, evidence (attachments, Controls, completed checklist items, gate decisions), last assessment date and assessor, and shall keep the history of assessments.
- **FR-DA-APL-05:** The system shall produce, for an Organization or a project, a Statement of Applicability and a compliance matrix (requirements × status) with coverage per standard and per clause, the list of gaps, and export to PDF, XLSX and CSV (Section 4.8).
- **FR-DA-APL-06:** The system shall alert the owner through the Notification Center when a requirement becomes Non-compliant, when an assessment is overdue, and when a new edition of an applied standard is published, and shall list the requirements affected by the new edition.
- **FR-DA-APL-07:** The system shall record the application, tailoring, waiver and assessment of every requirement in the audit trail, and shall never present a compliance status as a certification (FR-DA-CFG-09).

5. Common Non-Functional Requirements

5.1 Performance

- **NFR-DA-PERF-01:** Dashboard and list views shall render within 2 seconds under nominal per-tenant data volumes.
- **NFR-DA-PERF-02:** An AI agent suggestion shall return within 5 seconds under its deterministic path.
- **NFR-DA-PERF-03:** A knowledge-base or FAQ retrieval search shall return ranked results within 1 second for a tenant's full corpus.
- **NFR-DA-PERF-04:** A single standard-report export (PDF, Excel, or Word) shall complete within 3 seconds for a tenant's current data volume.
- **NFR-DA-PERF-05:** With 100 concurrent users on one tenant, the 95th-percentile response time of read endpoints shall stay at or below 1 second, measured by a repeatable load test. A per-user read cache, where used, shall hold an entry for no more than 30 seconds and shall be invalidated by any write in the same Organization.
- **NFR-DA-PERF-06:** Activating a vertical, with its processes and configuration, shall complete within 5 minutes (configurable target).
- **NFR-DA-PERF-07:** A bulk data import shall load 10,000 records within 30 minutes, and an SME shall be able to complete onboarding within 30 days (configurable targets).
- **NFR-DA-PERF-08:** A production deployment offering the Vertical and SME layers shall be sized for configurable targets, by default 1,000 API calls per second sustained and 10,000 concurrent users, with reports generated within 60 seconds.
- **NFR-DA-PERF-09:** Importing a standard of 5,000 requirements shall complete within 1 minute, and the compliance matrix of a project with 2,000 applicable requirements shall display within 3 seconds.

5.2 Security

- **NFR-DA-SEC-01:** All API endpoints except login shall require a valid, signed, time-boxed authentication token.
- **NFR-DA-SEC-02:** Passwords shall be stored only as salted cryptographic hashes; plaintext passwords shall never be logged or persisted.
- **NFR-DA-SEC-03:** Every authorization decision shall be enforced server-side; client-side permission and Pack-entitlement checks are a UX convenience only, never the security boundary.
- **NFR-DA-SEC-04:** No data query shall be constructed by concatenating untrusted input; all data access shall use parameterized statements or an equivalent safe-query mechanism.
- **NFR-DA-SEC-05:** Owner/Evaluator segregation of duties, and any equivalent server-enforced business rule, shall be enforced server-side, not merely hidden in the UI.
- **NFR-DA-SEC-06:** A one-Accountable-per-RACSI-activity rule, and any equivalent uniqueness rule, shall be enforced by a data-persistence-layer constraint, not merely by disabling an option in the UI.
- **NFR-DA-SEC-07:** A Real LLM Provider Connection's API key shall be captured and stored only client-side; the backend shall never persist it to any table, write it to any log line, or return it in any API response, and any server-side proxying of a forwarded key shall use it for exactly one outbound call and never write, log, cache, or hold it beyond that request.
- **NFR-DA-SEC-08:** Every quota check (Projects, OBS nodes, AI Use Cases, and any other Pack-bound dimension) shall be enforced at the point of creation/activation on the server, never solely by disabling a

client-side control; an over-quota request shall be rejected with HTTP 409, never silently truncated or partially applied.

- **NFR-DA-SEC-09:** Solution Pack entitlement shall be treated as an authorization axis independent of, and enforced in addition to, RBAC permissions; a route gated by both shall require both checks to pass, each independently testable and independently capable of producing a 403.
- **NFR-DA-SEC-10:** Individually identifiable records requiring restricted visibility (e.g. individual-level readiness or sentiment data) shall be restricted, by RBAC default, to an explicitly named set of roles; other roles shall see only aggregated, de-identified views.
- **NFR-DA-SEC-11:** Tenant data isolation shall be verified by an automated cross-tenant access test suite exercised in the build/release pipeline, not solely by code review.
- **NFR-DA-SEC-12:** The login endpoint shall be rate-limited per client, and every response shall carry standard security headers (content-type sniffing protection, frame protection, and a restrictive cross-origin resource policy).
- **NFR-DA-SEC-13:** Caches, backups, and logs shall never make one tenant's data available to another; backups shall be stored outside every web-served directory and be readable only by the service account and the platform administrator.
- **NFR-DA-SEC-14:** The system shall support standard federated authentication (SAML 2.0, OAuth 2.0, OpenID Connect) and multi-factor authentication.
- **NFR-DA-SEC-15:** Role-based access control shall support scopes by vertical and by SME track in addition to the tenancy scopes.
- **NFR-DA-SEC-16:** Data shall be encrypted at rest (AES-256 or equivalent) and in transit (TLS 1.2 or later), with configurable data residency regions and optional customer-managed keys (bring your own key).

5.3 Reliability & Data Integrity

- **NFR-DA-REL-01:** The system shall degrade gracefully if an AI agent or live-LLM call fails: the underlying CRUD operation shall still succeed, with the AI suggestion simply unavailable or served from its deterministic fallback.
- **NFR-DA-REL-02:** The system shall target 99.9% uptime in a production deployment.
- **NFR-DA-REL-03:** A report-export failure shall return a structured error without corrupting or partially writing the underlying report data view.
- **NFR-DA-REL-04:** The persistence layer shall support scheduled backup with a defined retention policy; a production deployment shall not rely on a single unreplicated data file as its sole durability boundary.
- **NFR-DA-REL-05:** The backup schedule shall bound data loss to at most 24 hours, and restoring a backup shall be tested at least once per release.
- **NFR-DA-REL-06:** A production deployment offering the Vertical and SME layers shall support disaster recovery with an RTO under 4 hours and an RPO under 1 hour, automatic failover and automated backups, tightening NFR-DA-REL-05 for that deployment.

5.4 Scalability

- **NFR-DA-SCALE-01:** The architecture shall support horizontal scaling of the web tier independently of the API tier.
- **NFR-DA-SCALE-02:** The system shall support at least 100 concurrent users per tenant without functional degradation.

- **NFR-DA-SCALE-03:** The persistence schema and the retrieval index shall each be swappable for a distributed equivalent (e.g. a managed relational database, a vector database) without changing the API contract.
- **NFR-DA-SCALE-04:** The system shall scale to configurable limits, by default 1,000 tenants per deployment including 10,000 SME tenants in shared mode, 100 million records, 500 macro processes, 200 E2E processes and 100 checklist items per gate.
- **NFR-DA-SCALE-05:** Adding a vertical, an SME track, a template, a gate or a checklist shall require no downtime.
- **NFR-DA-SCALE-06:** The catalogs shall support at least 500 standards and 100,000 requirements per tenant without functional degradation.

5.5 Usability & Accessibility

- **NFR-DA-UX-01:** The UI, and every generated document, shall follow the suite's common visual identity (color palette, typography, iconography) consistently across all modules and output formats.
- **NFR-DA-UX-02:** The UI shall be responsive and usable on desktop, tablet, and mobile viewport widths, with touch-target sizing meeting WCAG 2.1 AA on mobile.
- **NFR-DA-UX-03:** All interactive controls shall be keyboard-navigable and carry accessible labels (WCAG 2.1 AA target).
- **NFR-DA-UX-04:** Status, severity, and classification fields shall use one consistent color/badge vocabulary across every module, so a user who has learned a meaning in one module recognizes it in every other.
- **NFR-DA-UX-05:** Every interactive target shall measure at least 24×24 CSS pixels (WCAG 2.2 SC 2.5.8), and at least 44×44 CSS pixels on touch screens.
- **NFR-DA-UX-06:** Every release shall pass an automated accessibility scan (e.g. axe-core) of its main screens with no critical or serious violation. Text shall meet WCAG 2.1 AA contrast (4.5:1, or 3:1 for large text); where a brand color falls short, its darker tone shall be used for text, and any remaining brand conflict shall be recorded in Appendix A.
- **NFR-DA-UX-07:** Every confirmation, warning, and error message shown as a transient notification shall be translated (FR-DA-I18N-07), announced to assistive technologies through a polite live region, and remain readable long enough to act on.
- **NFR-DA-UX-08:** Wide matrices and tables shall scroll inside their own container, keeping the first column visible, so the page never scrolls horizontally on a 390-pixel viewport.
- **NFR-DA-UX-09:** SME users shall complete onboarding within 1 day and need less than 4 hours of training, helped by guided workflows (configurable targets).
- **NFR-DA-UX-10:** User satisfaction shall be measured, with a configurable target (default NPS of 30 or more).

5.6 Localization

- **NFR-DA-I18N-01:** Every layout — navigation, tables, and charts — shall render correctly mirrored under Arabic RTL, not just with mirrored text direction.
- **NFR-DA-I18N-02:** The system shall support configurable languages (default English, French, Arabic, Spanish, German), currencies (default USD, EUR, MAD, ZAR) and regional date, time and number formats.

5.7 Portability / Installability

- **NFR-DA-PORT-01:** Installation shall succeed on a fresh target machine with only a supported Node.js runtime installed — no native C/C++ build toolchain, no required external database server, and no mandatory external API key.
- **NFR-DA-PORT-02:** The same source shall run unmodified across supported operating systems via standard package-manager install/build/start commands.
- **NFR-DA-PORT-03:** Every change shall be built by a continuous-integration pipeline on Linux, Windows, and macOS that installs dependencies, seeds the demonstration data, runs the automated tests — including the cross-tenant suite of NFR-DA-SEC-11 — and builds the web client.
- **NFR-DA-PORT-04:** Each tier shall install, seed, and start with its package manager's standard commands (install, seed, dev or start), documented in a README at the root of the delivery.

5.8 Maintainability / Extensibility

- **NFR-DA-MAINT-01:** Cross-cutting engines — the Action/Task Register, the Alert & Notification Engine, the RBAC layer, the KPI/Reporting engine, and the Audit Trail Service — shall be implemented as composable modules reusable by sibling Dynamic Apps via the same API contract.
- **NFR-DA-MAINT-02:** AI agent implementations shall be isolated behind a stable internal interface so a rule-based implementation can be replaced by an LLM/vector-DB-backed one without changing calling code.
- **NFR-DA-MAINT-03:** The Solution Pack feature and quota matrix shall be defined in exactly one place and consumed by every enforcing route, so entitlements can never drift between routes.
- **NFR-DA-MAINT-04:** The AI Use Case Library's generation service shall be the single designated server-side integration point for a live LLM call and its provider-specific credential/endpoint logic; no other module shall embed its own LLM provider client.
- **NFR-DA-MAINT-05:** The alert type-to-name and type-to-stage/process mapping shall be defined in exactly one shared module, consumed by both seed data and live computation.
- **NFR-DA-MAINT-06:** The Installation Guide and User Guide shall be generated from source in Word and PDF, each with an automatic table of contents, and the User Guide's step-by-step walkthroughs shall be replayed automatically against a freshly seeded instance, so the documentation cannot drift from the application.
- **NFR-DA-MAINT-07:** Every User Guide walkthrough shall start from the tenancy — create or choose the Group (Yes/No), create the Organization, then the principal record(s) — and the guide shall include the decision matrix of FR-DA-HLP-04.
- **NFR-DA-MAINT-08:** Verticals, SME tracks, complexity criteria, packs, project templates, gates and checklists shall be configuration data loaded without code changes, and the platform shall be modular with zero-downtime upgrades, monitoring, alerting and comprehensive logging.

5.9 Compatibility

- **NFR-DA-COMPAT-01:** The application shall target evergreen desktop, tablet, and mobile browsers; no legacy-browser support is in scope.
- **NFR-DA-COMPAT-02:** BPMN diagrams shall be stored and exchanged as schema-valid BPMN 2.0 XML, verified by a round-trip import/export, so a diagram authored in one Dynamic App opens without error in any other BPMN 2.0-compliant tool, and vice versa.
- **NFR-DA-COMPAT-03:** The application's core workflows (record CRUD, approvals, alerts) shall remain fully usable on a mobile viewport without requiring a separate native application; a native or installable Progressive Web App wrapper, where offered, shall consume the same REST API (Section 7.2) with no mobile-specific endpoint variant.

- **NFR-DA-COMPAT-04:** Standards and requirements shall be exchanged in CSV, XLSX and ReqIF 1.2, verified by a round-trip import and export that preserves structure, identifiers, types and links.

5.10 Auditability & Compliance

- **NFR-DA-COMP-01:** Every AI-generated suggestion shall be logged with a confidence indicator and be attributable to a specific agent/use case and record, or, for the AI Assistant, to the requesting user and question.
- **NFR-DA-COMP-02:** The Controls module shall support demonstrating COSO Internal Control — Integrated Framework coverage across all five components.
- **NFR-DA-COMP-03:** The system shall retain the full version history of every AI Use Case and every template indefinitely (no hard deletion of prior versions on edit or revert).
- **NFR-DA-COMP-04:** The system shall never represent, in any UI text, Help content, FAQ answer, or generated document, that activating a Compliance & Security Standard constitutes a certification, an external audit, or a legal attestation of compliance.
- **NFR-DA-COMP-05:** The system shall let verticals map to sector standards — for example IATF 16949 (automotive), AS9100 (aerospace), EU MDR and ISO 13485 (medical devices), FDA 21 CFR Part 11 (pharmaceutical), HACCP (food safety) — as scaffolds subject to FR-DA-CFG-09; certification (ISO/IEC 27001, SOC 2, GDPR) is an attribute of a deployment, not of the software.
- **NFR-DA-COMP-06:** Every requirement of a Dynamic App shall be traceable to its business requirement, user story, test case, release and, where applicable, customer commitment, in line with ISO/IEC/IEEE 29148; requirement changes shall be classified Minor, Moderate, Major or Breaking with the matching approval and regression scope.
- **NFR-DA-COMP-07:** The system shall store the full text of a copyrighted standard only under a licence recorded with it, and otherwise only its clause references, titles and paraphrased requirements.
- **NFR-DA-COMP-08:** Requirements in the catalog shall follow the characteristics of ISO/IEC/IEEE 29148 (necessary, unambiguous, verifiable, singular), and the system shall flag on entry a requirement without an obligation level or verification method.

5.11 Data Retention & Privacy

- **NFR-DA-DATA-01:** By default, and whenever no Real LLM Provider Connection is configured, the system shall not transmit tenant data to any third-party service; every AI capability shall run entirely within the deployment's own infrastructure using rule-based/statistical methods and RAG grounding. Only an explicitly user-configured, client-stored Real LLM Provider Connection may cause a single generation request to be sent to that user-selected third-party provider — opt-in per user, never enabled by default, and never applied to a permission-scoped data-query mode.
- **NFR-DA-DATA-02:** The system shall support both shared-infrastructure (SaaS) and dedicated/sovereign deployment options per Organization, without changing the tenant-isolation guarantee.

6. Common Data Model

The following core entities recur, under equivalent names, across the Dynamic Apps Suite. Each entity carries a tenant (organization_id) foreign key except Group and any global, platform-wide catalog.

Entity	Description
Group	Optional holding entity above Organization; groups sibling Organizations.
Organization	The tenant boundary; holds sector/country/profile attributes and all tenant-isolated data.
Project	Optional sub-entity of an Organization; the principal record type may reference one; count enforced against the active Pack's quota.
ObsNode	A node in the OBS tree; referenced by users and by every governance entity below; count enforced against the active Pack's quota.
User / Role / Permission	Person, named role, and a global permission catalog entry (module.action) granted to roles.
License / Configuration	Configuration Management record: Solution Pack, Add-On flags, Compliance Standard flags, deployment option, support tier, seats, billing.
GovernanceSettings	KPI thresholds, alert toggles, and process-default/justification-requirement settings per Organization.
Standard / StandardRequirement	A reference standard and its clause-level requirement breakdown, each illustrated with an example.
Principal Record (e.g. NCPFiche / CM Project)	The central aggregate the application manages through its defined multi-stage lifecycle.
Action / ActionEvaluation / ActionEvidence	Abstract action base, its Owner/Evaluator-segregated effectiveness verdict, and supporting evidence attachments.
NotificationAlert	An instance of a cataloged alert type, tagged with its process/stage.
AIAgentLog / AIUsageLog	Immutable logs of every AI agent invocation and every AI suggestion's accepted/edited/rejected outcome.
AuditLog	Immutable log of every create/update/delete on a governed entity.
AIUseCase / AIUseCaseVersion / AIUseCaseProjectOverride	Governed AI Use Case metadata, its versioned content, and its per-Project tri-state activation override.
BusinessRule / Control / RiskOpportunity	Typed governance records, each optionally tagged to an OBS node, a process/stage, and (Control) a Compliance Standard.
RacsiActivity / RacsiAssignment	A governed activity and its R/A/C/S/I assignments, at most one Accountable per activity.
BpmnDiagram	A stored BPMN 2.0 process diagram (XML payload), owned by an OBS node.
CustomKpi	An organization-defined KPI additive to the platform's built-in KPI set.
Template (Sheet/Charter/Communication/Training, etc.)	A versioned, pre-filled starting point for the application's principal record type or a related artifact.
WbsNode	A node in a Work Breakdown Structure, referencing one or more Action IDs as leaf-level work items, with planned dates, percent-complete, and dependency links to other WBS nodes/Actions.
EntityVersion	A generic, immutable version record produced by the Version Management service for any entity type registered with it; carries the entity's full state at that version, the acting user, and a timestamp.
CommunicationDispatch /	A composed message routed through one or more channels (in-app, email, SMS,

Entity	Description
DeliveryStatus	push, webhook) and its per-channel delivery status (queued/sent/delivered/failed).
RexEntry	A Return on Experience record captured at the closure of a principal record or an Action: what went well, what didn't, root cause, recommendation, category/tag, and an effectiveness rating; versioned and indexed into the retrieval corpus.
ExternalIntegration	A registered connection to an external application (CRM, ERP, calendar, storage, identity provider, etc.), its own enable/disable state, Solution-Pack/Add-On gate, credentials reference, and connection-health/audit log.
UserPreference	Per-user settings stored server-side: display language, menu dock position, pinned state, collapsed menu groups, favorites, and notification channel choices.
BenchmarkMetric / BenchmarkSharing	A catalog entry of a comparable metric (unit, direction of good, computation), and an Organization's Group-sharing setting, audited on every change.
Backup	A dated, consistent copy of the data store, with its size and retention expiry; listed to the platform administrator only.
RequirementTrace	A requirement ID of this standard with its implementation status and evidence note, shown in the traceability view.
ObsMember	A person placed in an OBS node, with the standard role and the role played there.
ChecklistTemplate	A versioned list of checklist items in the Checklist Library, universal or specific to a vertical, mode or track, associated with zero or more gates (GateChecklist).
Attachment	A file attached to an action or a checklist item: name, type, size, author, date, tenant-scoped storage reference.
Vertical	A sector entity: prefix, name, taxonomy parent, lifecycle state, owner, version; links to its macro and E2E processes, data extensions, compliance mappings and configuration.
VerticalActivation	The activation of a vertical by an Organization, with its version, validation result and activation order.
SmeTrack	An SME rigor level: processes, gates, checklist items per gate, typical duration, lifecycle state.
ComplexityCriterion	A scoring criterion and its weight, universal or specific to one vertical, versioned.
ComplexityScore	The scored criteria of a project, the result, the recommended and chosen track, and any override justification and approval.
ProjectTemplate	A catalog entry: scope (universal or vertical), mode and track, status, version, fields, phases with their attached gates and checklists, roles and milestones.
GateDefinition	A Gate Library entry: criteria, approvers, decision options, applicability, version.
GateChecklist	The association of a checklist template with a gate, optionally per track, with its order and whether it is mandatory.
PhaseAttachment	The attachment of a gate, or of a checklist not related to a gate, at a given version, to a phase of a project or a template.
Pack	An SME pack, vertical pack or bundle: contents, target segment, pricing rules.
OnboardingPlan	The onboarding of an SME: steps, imports, validation results, dates, metrics.
Standard (catalog)	A Standards Catalog entry: code, title, category (Internal or External), kind, issuer, edition, dates, status, owner, licence, ownership level (platform or Organization).
StandardNode	A node of a standard's structure — part, chapter, clause or sub-clause — with number, title, guidance, parent and order.
Requirement	A Requirements Catalog entry: identifier, title, statement, type, obligation, verification method, expected evidence, source, owner, tags, status, version; optionally placed at a StandardNode.

Entity	Description
RequirementType	A configurable requirement type with its active flag.
RequirementLink	A typed link between two requirements (derives from, refines, satisfies, equivalent to, conflicts with), or between a requirement and a Control, checklist item, gate criterion, process, task or attachment.
Applicability	A standard, node or requirement applied to an Organization and/or a project, with status, justification, waiver and inheritance source.
ComplianceAssessment	A dated assessment of an applicable requirement: status, evidence, assessor, comments.

7. Common External Interface Conventions

7.1 User Interface Conventions

Every screen is accessible only when the authenticated user holds the corresponding view permission and, where applicable, the required Solution Pack entitlement. A Dashboard, a Configuration Management screen, a Permission Matrix screen, a Help screen, an Alerts inbox, a Reports screen with per-format download actions, and an AI Assistant (dedicated screen plus floating widget) are present in every Dynamic App.

Every screen sits inside the common navigation shell (Section 4.19). A Benchmarking screen, with Within the organization and Across the group views, is present in every Dynamic App that holds comparable records (Section 4.20), and a Backups panel is present on the platform administrator's configuration screen (Section 4.21). A screen the user may not open shows an access notice (FR-DA-RBAC-08) rather than an empty page.

Revision 1.2 adds a tenancy screen (Group Yes/No > Organization > records), a portfolio overview, a checklist template library in the process design, an attachments panel on every action, AI badges on the process design, and a live AI model card in personal settings and in administration.

Revision 1.3 adds screens for verticals and their activation, SME tracks and complexity criteria, packs, the project template catalog, a project creation entry with three tabs (From the catalog, Manual, With AI), a Checklist Library (checklists related to a gate or not) and a Gate Library (gates with their associated checklists), both attachable to phases.

Revision 1.4 adds a Standards Catalog (list and structure tree), a Requirements Catalog with types and links, a Standards & Requirements tab on each Organization and project, and a compliance matrix with its Statement of Applicability.

7.2 Software Interfaces (REST API)

All interfaces are exposed as a JSON REST API, secured by a bearer token (except the login endpoint), and enforced per-endpoint by the RBAC permission catalog and, where applicable, Solution Pack entitlement middleware. At minimum, every Dynamic App exposes equivalents of the following endpoint families:

- Auth: login, current-user profile, language preference update.
- Hierarchy & OBS: Group/Organization/Project CRUD, OBS CRUD, quota-checked creation.
- Users, Roles & Permissions: user CRUD, role assignment, permission-matrix read/write.
- Configuration Management: read effective configuration and static catalog (open to any authenticated user); write Pack/Add-On/Compliance selection (permission-gated); read live usage against quota.
- Governance: Business Rules, Controls, Risks & Opportunities, RACSI, Custom KPIs — each with process/stage filtering.
- AI Use Cases: catalog CRUD, activation, per-Project override, version history and revert, append-only usage log.
- BPMN: diagram CRUD, view requiring at least view-mode entitlement, write requiring full-edit entitlement.
- AI Assistant: a data-query endpoint and an application-help endpoint, each Pack-gated.

- Reports: per-report data endpoints and a per-report export endpoint accepting a target format (pdf/xlsx/docx).
- Alerts: list, run/recompute, mark-read.
- User preferences: read and update the current user's navigation and display preferences.
- Workflow steps: complete, and reopen with justification (FR-DA-AUD-05).
- Benchmarking: metric catalog, within-organization comparison, Group comparison, and read/update of the Organization's sharing setting.
- Operations: health (unauthenticated, no tenant data); backup list and on-demand backup (platform administrator).
- Tenancy and portfolio: tenancy tree, scope lists (groups, organizations, records the user may see), records across the scope, portfolio matrix.
- Checklist templates: library CRUD with version history; add to a gate or phase checklist from a template or by hand; save it as a template.
- Attachments: accepted types, multi-file upload, download, delete.
- AI model: model and provider catalog; connection test.
- Verticals: CRUD, lifecycle transitions and approvals, versions and rollback, activation with validation, deactivation with dependency check.
- SME and scoring: tracks CRUD, complexity criteria and weights, score computation, track override with approval.
- Project templates: catalog list with filters, CRUD, duplicate, publish and retire, version history, save a project as a template.
- Project creation: create from a template, manually, or from an AI draft (draft, then accept with the retained items).
- Checklist Library: CRUD, duplicate and versions; filter by gate or "not related to a gate"; gates using a checklist; attach a stand-alone checklist to a phase, task or project.
- Gate Library: CRUD, duplicate and versions; associate, order and detach checklists; attach, reorder and detach gates on a phase; decisions and waivers.
- Packs and onboarding: pack and bundle CRUD with pricing rules, upgrade and downgrade; onboarding plan, import and validation.
- Standards: catalog CRUD, duplicate, lifecycle and editions with comparison; structure tree CRUD and reorder; import and export (CSV, XLSX, ReqIF, PDF); AI draft from a document.
- Requirements: catalog CRUD, bulk edit, versions; requirement types CRUD; links between requirements and to implementing objects; where-used; import and export; AI duplicate and link suggestions.
- Applicability and compliance: apply and unapply to an Organization or a project; tailoring and waivers; assessments with evidence; Statement of Applicability and compliance matrix export.

7.3 Hardware Interfaces

None. A Dynamic App is a standard web application with no direct hardware interface requirement.

7.4 Communications Interfaces

HTTP/HTTPS using JSON request/response bodies for all client-server communication, except report-export responses (application/pdf, .xlsx, .docx binary bodies with a Content-Disposition attachment header). Authentication is via a signed bearer token issued at login.

Appendix A — Conformance & Divergence Notes

This standard states the target platform behavior every Dynamic App should converge on. The two source applications compared to produce it are, as of their respective v2.0 SRS documents, not identical in how completely they satisfy it — recorded here for transparency rather than silently smoothed over.

Revision 1.1 adds a third reference, the CortexPLM build of September 2026, and rows for the capabilities this revision introduces. Where a 1.0 target requirement is now observed in CortexPLM, its row says so.

Area	Standard Requirement	Current Divergence
Server-side authorization (NFR-DA-SEC-01/03)	Full JWT authentication and per-endpoint RBAC + Pack-entitlement enforcement, server-side.	RBAC and Pack-gating (Configuration Management module) are enforced client-side only in the current reference build; the API accepts any caller with no credential check. Closing this gap is the single highest-priority item for a production-grade rebuild. Revision 1.1: CortexPLM enforces JWT authentication, per-endpoint RBAC, and Pack entitlement server-side, covered by automated tests.
Persistence model	Any relational or document store that preserves per-entity querying.	NCP Solver uses a normalized relational schema (node:sqlite). journi persists one whole-tenant JSON document per installation and overwrites it wholesale on every save, with no optimistic locking — acceptable for its single-machine reference context, not for concurrent multi-user production use.
Solution Pack model	Three Packs, priced Add-Ons, Compliance Standards as three independent axes.	Both applications implement this model; naming of the three Packs and the specific module list they unlock differs per product (Resolve/Govern/Assure vs. Trailhead/Waypoint/Horizon), which is expected — the axis structure, not the tier names, is the common requirement.
Compliance & Security Standards catalog	A defined set of independently toggleable standards.	NCP Solver ships three (GDPR, ISO/IEC 27001, SOC 2); journi ships seven (adds ISO/IEC 27701, HIPAA, FIPS 140-2, ISO 9001). The mechanism (Section 4.3) is common; the catalog's specific membership is product-specific and may extend beyond this standard's minimum.
AI Assistant chatbot	RBAC-scoped data query mode plus application-help mode, both Pack-gated.	Both applications implement this pattern; journi additionally composes a third, process-knowledge-base retrieval source into its chatbot (Section 4.5), which is a reasonable extension of, not a deviation from, FR-DA-AST-04.
Governance-artifact scoping	Business Rules/Controls/Risks/RACSI taggable to a process or stage.	NCP Solver tags to the S1–S7 Sheet stage directly; journi tags to its Macro Process catalog (a stable, platform-

Area	Standard Requirement	Current Divergence
		wide process taxonomy) rather than a per-project task instance. Both satisfy FR-DA-GOV-08 at the level of granularity each application's own process model supports.
WBS & Gantt Scheduling (Section 4.15)	Any grouping of Actions can be organized into a WBS and rendered as a Gantt chart.	New in this revision. Not present in either Journi v2.0 or NCP Solver v2.0; neither reference application's current build implements Action grouping, scheduling fields, or a Gantt renderer. Recorded here as a target requirement, not yet observed in the suite. Revision 1.1: implemented and verified in the CortexPLM reference build.
Generic Version Management (Section 4.16)	Any registered entity type shares one versioning service with full history and revert-by-duplication.	New in this revision, generalized from the AI Use Case Library and Template Libraries pattern both applications already implement (FR-DA-AI-03, FR-DA-TPL-02). Neither reference application currently exposes this as a shared service other entity types can register with; each still implements versioning per entity type. Revision 1.1: implemented and verified in the CortexPLM reference build.
Multi-Channel Communication (Section 4.17)	A notification is dispatched through one or more channels (in-app, email, SMS, push, webhook) behind a shared interface.	New in this revision. Both reference applications currently deliver notifications in-app only (Section 4.9); neither implements email, SMS, push, or webhook dispatch. Recorded here as a target requirement, not yet observed in the suite. Revision 1.1: implemented and verified in the CortexPLM reference build.
Mobile Compatibility (Sections 2.5, 5.5, 5.9)	Core workflows remain fully usable on a mobile viewport; a PWA/native wrapper, where offered, uses the same REST API.	New in this revision. Both reference applications currently target desktop/tablet only; neither has been verified against a mobile viewport or ships a PWA/native wrapper. Revision 1.1: CortexPLM verified at a 390-pixel viewport with no horizontal scrolling and 44-pixel touch targets; no native wrapper.
Return on Experience / REX (Section 4.18)	A REX entry is captured at closure of a Project or Action, versioned, and indexed for retrieval.	New in this revision. Neither Journi nor NCP Solver currently prompts for or stores a structured lessons-learned entry at closure; any post-mortem practice today is informal and outside the application. Recorded here as a target requirement, not yet observed in the suite. Revision 1.1: implemented and verified in the CortexPLM reference build.
External Integration Registry (Sections 3.3, 4.3)	An Organization can register, credential, health-check, and independently toggle a connection to an external application.	New in this revision. Both reference applications currently integrate with, at most, one fixed external system each, configured outside the product rather than through a generic, user-facing registry. Recorded here as a target

Area	Standard Requirement	Current Divergence
		requirement, not yet observed in the suite. Revision 1.1: implemented and verified in the CortexPLM reference build.
Navigation Shell & Personalization (Sections 3.11, 4.19)	One shell; menu pinned or sliding, docked on any edge; favorites and collapsed groups stored server-side; RTL mirroring; mobile drawer; keyboard operation.	New in revision 1.1. Implemented and verified in CortexPLM. Neither the journal nor the NCP Solver v2.0 SRS specifies menu personalization; to be assessed.
Internal Benchmarking (Sections 3.12, 4.20)	Metrics compared across segments of one Organization and across the Organizations of one Group, aggregates only, minimum sample, opt-out; no external benchmarking.	New in revision 1.1. Implemented in CortexPLM: 13 metrics by project type, track, and department; Group comparison with ranks, median, opt-out, and a minimum of 3 projects. Not present in journal or NCP Solver v2.0.
Platform Operations (Sections 3.13, 4.21)	Backups with retention, additive startup migrations, startup guards, health endpoint, deterministic seed, traceability view.	New in revision 1.1. Implemented in CortexPLM (daily backup, 14-day retention). Partial for FR-DA-OPS-05: the runtime version is declared in the package manifest and checked at install time, not by the server at startup. Off-site backup copies remain an operator task (Section 2.7).
Opaque identifiers (FR-DA-TEN-11)	Identifiers reveal nothing about other tenants; foreign records answer 404.	New in revision 1.1. Partial in CortexPLM: foreign records answer 404, but record ids are one global sequence, so an id's size can hint at overall platform activity.
Contrast of brand colors (NFR-DA-UX-06)	WCAG 2.1 AA text contrast; darker brand tone for text where needed.	New in revision 1.1. CortexPLM passes the automated scan except two brand-mandated pairs — white text on the primary orange (2.3:1) and orange key figures (3.0:1) — which conflict with the Brand & Graphical Chart Guide and are recorded here rather than changed.
Other 1.1 additions (TEN-09/10, RBAC-08–10, AUD-05, I18N-07/08, VER-06, REX-08, PERF-05, SEC-12/13, REL-05, UX-05/07, PORT-03/04, MAINT-06)	See Appendix B.	New in revision 1.1. All implemented and verified in CortexPLM (load test at 100 users: 95th percentile 0.9 s; CI on Linux, Windows, and macOS).
Tenancy, provisioning and group visibility (FR-DA-TEN-12 – 16)	Tenancy tree and table; organization chosen before a record; provisioning with starting team; read-only group visibility; OBS members with roles.	New in revision 1.2. Implemented and verified in CortexPLM (automated tests: group scope, independent scope, platform scope, provisioning with 24 accounts and duplicate-domain refusal).
Portfolio overview, checklist library, attachments (Sections 4.22 – 4.24)	Scope matrix with legend; stage checklist templates linked or added on demand; multi-format attachments.	New in revision 1.2. Implemented and verified in CortexPLM, including the User Guide walkthroughs replayed against a fresh installation.
AI placement and model selection (FR-DA-AI-12 – 15), BPMN workspace (FR-DA-BPMN-06 – 08)	Tier badges runnable in place; custom use case delete; model list with custom choice and connection test; palette section and full screen.	New in revision 1.2. Implemented in CortexPLM. The connection test was verified for the authentication-failure outcome; a live call needs the user's own key.

Area	Standard Requirement	Current Divergence
Verticals and SME layers (Sections 4.25 – 4.32)	Verticals as governed configuration entities; SME mode with tracks; weighted complexity scoring; onboarding; packs and bundles.	New in revision 1.3. Target. CortexPLM covers part of it: an Organization carries a sector with seeded templates and checklists, and seven fixed criteria recommend a Full, Light or Fast track, with written justification for another choice. Configurable weights, vertical lifecycle and activation, SME tracks, onboarding and packs are not implemented.
Project Template Catalog and creation modes (Sections 4.33 – 4.34)	Catalog classified universal or per vertical, Full or SME; CRUD and versions; creation from the catalog, manually or with AI.	New in revision 1.3. Partly implemented in CortexPLM: project templates with CRUD and versions, creation from a template or manually, and AI suggestions on the project. Scope and mode classification, publication states and a full AI-drafted project are not implemented.
Checklist Library and Gate Library (Sections 4.23 and 4.35)	Checklists as independent entries, related to a gate or not; gates as library entries with CRUD and associated checklists; both attachable to any phase.	Reworded or new in revision 1.3. CortexPLM keeps checklist templates per gate and track, copied when the gate opens, and adds items from a template or by hand; stand-alone checklists, gate CRUD, gate-checklist association and attachment to any phase are a target, because its gates are fixed per track in code.
Standards & requirements management (Sections 4.36 – 4.38)	Standards Catalog from chapter to requirement; typed Requirements Catalog; applicability to Organizations and projects; compliance tracking and Statement of Applicability.	New in revision 1.4. Target. CortexPLM covers part of it: GDPR, ISO/IEC 27001 and SOC 2 are toggled per Organization and seed tagged Controls, and a reference knowledge library supports AI grounding. The catalogs, the structure tree, requirement types and links, and project-level applicability are not implemented.

A future revision of any product's SRS that closes one of the gaps above should update this appendix in the same revision, so the standard and the observed state of the suite never drift silently apart.

Appendix B — Revision 1.1 Change Log

Revision 1.1 is additive. Every 1.0 requirement keeps its ID and wording, except the editorial changes marked Clarified below. New requirements take the next free number in their section, so a conformance record written against 1.0 remains valid against 1.1.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.5, 1.6	Version 1.1; CortexPLM reference	Version raised to 1.1. CortexPLM added as a third reference; purpose and overview explain the revision.
1.2 Scope	Added	Navigation shell, internal benchmarking, and platform operations brought into scope; external benchmarking stated out of scope.
1.4 Definitions	6 terms added	Navigation Shell, Pinned / Sliding Menu, Dock Position, Internal Benchmarking, Minimum Sample Size, Additive Migration.
2.2, 2.3	3 principles, 3 functions	Personal navigation shell; compare within the Group only; upgrade and recover in place.
2.5 Operating Environment	Clarified	Recommended Node.js raised from 22.5+ to 22.13+ (built-in SQLite without an experimental flag); required version declared in the package manifest.
2.6, 2.7	1 constraint, 1 assumption	Benchmark data leaving a tenant is aggregated only; off-site backup copies are the operator's responsibility.
3.1 Tenancy	Clarified	Group membership is the benchmarking boundary; Group Yes/No and Independent shown on hierarchy screens.
3.11 – 3.13	New sections	Navigation Shell & Personalization, Internal Benchmarking, and Platform Operations architectures.
4.1 Tenant & Hierarchy	FR-DA-TEN-09 – 11	Group membership display; same-tree OBS parents; opaque identifiers and 404 for foreign records.
4.2 Identity & RBAC	FR-DA-RBAC-08 – 10	Access notice instead of a blank page; restricted evaluation-content permission; additive permission migration at startup.
4.11 Audit Trail	FR-DA-AUD-05	Reopen a completed step, with justification, before its stage is decided.
4.14 Localization	FR-DA-I18N-07 – 08	Server messages and assistant answers translated; multilingual Knowledge Base seed with language filter.
4.16 Version Management	FR-DA-VER-06	One shared side-by-side version comparison with changed fields highlighted.
4.18 REX	FR-DA-REX-08	REX prompt shown to whoever closes the record, with Skip that keeps the closure.
4.19 Navigation Shell	FR-DA-NAV-01 – 10 (new)	Menu from one navigation model; collapsible groups; pin / slide away; dock left, right, top, or bottom; favorites; server-side preferences; RTL mirroring; mobile drawer; keyboard operation; standard header bar.
4.20 Internal Benchmarking	FR-DA-BMK-01 – 09 (new)	Metric catalog; within-organization segments; Group ranks with ties; aggregates only; minimum sample; opt-out with audit; permissions; no external benchmarking; charts with filters.
4.21 Platform	FR-DA-OPS-01 – 08	On-demand and daily backup; retention; restore;

Section	Requirement IDs / Type	Change
Operations	(new)	additive migrations; startup guards; health endpoint; deterministic seed; traceability view.
5.1 Performance	NFR-DA-PERF-05	95th-percentile read time of 1 s or less at 100 concurrent users; rules for a per-user read cache.
5.2 Security	NFR-DA-SEC-12 – 13	Login rate limiting and security headers; no cross-tenant exposure through caches, backups, or logs.
5.3 Reliability	NFR-DA-REL-05	Data loss bounded to 24 hours; restore tested each release.
5.5 Usability & Accessibility	NFR-DA-UX-05 – 07	Target sizes 24 px / 44 px; automated accessibility scan and contrast rule; accessible, translated notifications.
5.7 Portability	NFR-DA-PORT-03 – 04	CI on Linux, Windows, and macOS with seed, tests, and build; standard install / seed / start commands.
5.8 Maintainability	NFR-DA-MAINT-06	Guides generated in Word and PDF with a TOC; walkthroughs replayed against a fresh seed.
6 Data Model	4 entities	UserPreference, BenchmarkMetric / BenchmarkSharing, Backup, RequirementTrace.
7.1, 7.2 Interfaces	Clarified	Shell, Benchmarking screen, Backups panel, and access notice; endpoint families for preferences, workflow steps, benchmarking, and operations.
Appendix A	Updated	CortexPLM status added to six 1.0 target rows and server-side authorization; six rows added for 1.1 capabilities; closing note widened to any product.

Nothing was removed. Product-specific material from CortexPLM — its phase-gate tracks, gate criteria, industry content, and commercial packaging — was deliberately left out, because it does not apply to every Dynamic App.

Appendix C — Revision 1.2 Change Log

Revision 1.2 is additive, like 1.1: every earlier requirement keeps its ID and wording, and new requirements take the next free number in their section.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.6	Version 1.2	Version raised to 1.2; purpose and overview describe the revision.
1.4 Definitions	4 terms added	Tenancy Tree, Provisioning, Portfolio Overview, Stage Checklist Template.
4.1 Tenant & Hierarchy	FR-DA-TEN-12 – 16	Tenancy tree and table; organization chosen before creating a record; provisioning of new organizations with an optional starting team; read-only visibility within the group; OBS members with roles.
4.4 AI Use Case Library	FR-DA-AI-12 – 15	Tier badges shown and runnable on the process design; links to step, process, task or E2E process; delete for custom use cases; model list with custom choice, custom endpoint and connection test.
4.7 BPMN	FR-DA-BPMN-06 – 08	Shapes palette section (legend when view-only); full-screen workspace with zoom and return; statement of whether diagrams drive execution.
4.12 Help	FR-DA-HLP-04	Decision matrix next to scored inputs; legend for every code or colour vocabulary.
4.19 Navigation	FR-DA-NAV-11	Menu organized by what users do, in a documented order.
4.21 Operations	FR-DA-OPS-09	Demonstration seed covers every module.
4.22 Portfolio Overview	FR-DA-PFO-01 – 04 (new)	Scope, record × stage matrix, legend and totals, CSV export, read-only foreign records.
4.23 Stage Checklist Library	FR-DA-CHK-01 – 05 (new)	Versioned templates per stage and variant; linked templates copied at stage opening; add from template or by hand without duplicates; sector templates; frozen after submission.
4.24 Attachments	FR-DA-ATT-01 – 04 (new)	Multi-file upload; published accepted formats with limits; metadata, download, audited removal; storage and backup.
5.5 Usability	NFR-DA-UX-08	Wide matrices scroll in their own container; no page-level horizontal scroll on phones.
5.8 Maintainability	NFR-DA-MAINT-07	User Guide walkthroughs start from the tenancy and include the decision matrix.
6 Data Model	3 entities	ObsMember, ChecklistTemplate, Attachment.
7.1, 7.2 Interfaces	Clarified	New screens and endpoint families for tenancy and portfolio, checklist templates, attachments and the AI model.
Appendix A	3 rows	Conformance of the 1.2 additions in CortexPLM.

Product-specific material of this CortexPLM round — its sectors, its phase-gate vocabulary and its demonstration content — stays in the CortexPLM documentation.

Appendix D — Revision 1.3 Change Log

Revision 1.3 keeps every earlier requirement ID. It is additive except in Section 4.23, where five requirements are reworded so that checklists and gates are separate libraries; new requirements take the next free number in their section or a new family.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.5, 1.6	Version 1.3	Version raised to 1.3; purpose and overview describe the revision; the Verticals and SME source added to the references.
1.4 Definitions	11 terms added, 1 reworded	Stage Checklist Template renamed Checklist Template. Added: Vertical, Vertical Instance, SME, Mode (Full / SME), SME Track, Complexity Score, Project Template Catalog, Creation Mode, Phase, Gate Definition, Pack / Bundle.
2.3 Product Functions	1 function added	Verticals, SME mode, complexity scoring, template catalog, creation modes, gates and checklists on any phase.
3.14 (new)	Architecture	Five requirement layers; verticals and tracks as configuration data; ordered, additive activation.
4.13 Template Libraries	FR-DA-TPL-05	Project templates are managed through the catalog of Section 4.33.
4.23 Checklist Library (renamed)	FR-DA-CHK-01 – 05 reworded; CHK-06 – 07 (new)	Renamed from Stage Checklist Library. A checklist is an independent library entry, related to zero, one or several gates; stand-alone checklists attach to a phase, task or project without blocking progression; impact shown before change or deletion.
4.25 Vertical Framework & Governance	FR-DA-VRT-01 – 10 (new)	Definition, taxonomy, extensibility, lifecycle and approval, versioning and rollback, tenant isolation, guarded deactivation, audit, ownership and change management, metrics.
4.26 Vertical Processes	FR-DA-VPR-01 – 09 (new)	Macro processes (3–5) and E2E processes (2–4) per vertical; taxonomy, compliance mapping, tasks and steps, chains, triggers, terminal states, metrics and reports.
4.27 Vertical Data, Compliance & Integration	FR-DA-VDT-01 – 03, VCP-01 – 03, VIN-01 – 02 (new)	Data model extension, exchange formats, retention and archiving; compliance requirements, monitoring, audits and training; connectors, security, monitoring and error handling.
4.28 Vertical Configuration	FR-DA-VCF-01 – 04 (new)	Configurable parameters, pre-built templates, validation before activation, versioning and audit.
4.29 SME Mode & Tracks	FR-DA-SME-01 – 08 (new)	SME definition and segments, cross-sector activation, 2–4 tracks, SME processes, extensibility, assignment and override, metrics, governance.
4.30 Complexity Scoring	FR-DA-SCO-01 – 04 (new)	Weighted configurable criteria summing to 100 %, vertical drivers, audit and reporting.
4.31 SME Onboarding & Experience	FR-DA-ONB-01 – 05 (new)	30-day onboarding plan, import and validation, guided training, simplified mobile experience and self-service support, metrics.
4.32 Packaging & Pricing	FR-DA-PKG-01 – 04 (new)	SME packs and vertical bundles as Solution Pack variants, pricing rules, upgrade and downgrade, reports.
4.33 Project Template	FR-DA-PTC-01 – 06	Universal or vertical, Full or SME templates; CRUD,

Section	Requirement IDs / Type	Change
Catalog	(new)	duplicate and versions; catalog browsing; Draft, Published, Retired; seeding and save-as-template; permissions.
4.34 Project Creation Modes	FR-DA-PCM-01 – 05 (new)	From the catalog, Manual, With AI (reviewed item by item); scoring, quotas and audit in every mode.
4.35 Gate Library	FR-DA-GTE-01 – 07 (new)	Gate definitions with CRUD and versions; association of checklists from the Checklist Library; attachment of gates to any phase; vertical- and track-aware gates; enforcement and waivers; decisions and audit; library changes never alter started phases.
5.1, 5.2, 5.3, 5.4	NFR-DA-PERF-06 – 08, SEC-14 – 16, REL-06, SCALE-04 – 05	Activation, import and production sizing targets; federated authentication and MFA, vertical and track scopes, encryption, residency and BYOK; disaster recovery; scale limits and zero-downtime additions.
5.5, 5.6, 5.8, 5.10	NFR-DA-UX-09 – 10, I18N-02, MAINT-08, COMP-05 – 06	SME onboarding and training targets, satisfaction; languages, currencies and formats; configuration-driven layers; sector standards scaffolds; requirement traceability and change classes.
6 Data Model	11 entities, 1 reworded	Vertical, VerticalActivation, SmeTrack, ComplexityCriterion, ComplexityScore, ProjectTemplate, GateDefinition, GateChecklist, PhaseAttachment, Pack, OnboardingPlan; ChecklistTemplate no longer tied to one stage.
7.1, 7.2 Interfaces	Clarified	New screens and endpoint families for verticals, SME and scoring, templates, creation modes, gates and checklists, packs and onboarding.
Appendix A	3 rows	Conformance of the 1.3 additions in CortexPLM.
Appendix E (new)	Traceability	Mapping of every source requirement family to the requirements of this standard.

Appendix E — Traceability to the Verticals and SME Requirements

Each requirement family of the Application-Agnostic Requirements for Verticals and SME source is covered as follows. The 29 vertical instances of the source (REQ-VERT-{SECTOR}-...: Automotive, Aerospace & Defense, Industrial Machinery, Healthcare & Life Sciences, Electronics & High-Tech, AEC & Construction, Food & Beverage, Social & Solidarity Economy, Pharmaceutical & Process, Retail, Footwear & Apparel, Consumer Packaged Goods, Energy & Utilities, Telecommunications, Process/Petrochemical, Shipbuilding, Consumer Products Extended, Process Industries, Infrastructure, Fabrication & Assembly, Process/Packaged Goods, Semiconductor & Electronics, Retail, Distribution, Logistics, Pharma, Manufacturing (Generic), Private Education, Public Education, Healthcare Providers) are content delivered as Vertical Instances under Sections 4.25 to 4.28; they stay in the vertical packs, not in this standard.

Source family	Requirements in this standard	Note
REQ-VERT-FW-001 – 010	FR-DA-VRT-01 – 08	FW-005 and FW-008 merged into VRT-04 and VRT-05.
REQ-VERT-GOV-001 – 005	FR-DA-VRT-04, 08, 09, 10	GOV-002 and GOV-004 duplicate FW-005 and FW-010.
REQ-VERT-MP-001 – 010	FR-DA-VPR-01 – 05, 09	Tasks, steps and dependencies merged into VPR-05.
REQ-VERT-E2E-001 – 007	FR-DA-VPR-06 – 09	Triggers and terminal states merged into VPR-07.
REQ-VERT-DATA-001 – 005	FR-DA-VDT-01 – 03	Retention and archiving merged into VDT-03.
REQ-VERT-COMP-001 – 005	FR-DA-VCP-01 – 03, NFR-DA-COMP-05	Kept subject to FR-DA-CFG-09.
REQ-VERT-INT-001 – 005	FR-DA-VIN-01 – 02	Builds on FR-DA-CFG-10 – 14.
REQ-VERT-CONF-001 – 005	FR-DA-VCF-01 – 04	Audit merged into VCF-04.
REQ-SME-FW-001 – 005	FR-DA-SME-01, 02, 05	
REQ-SME-MP-001 – 005, REQ-SME-E2E-001 – 005	FR-DA-SME-04	Reuses the vertical process rules.
REQ-SME-TRACK-001 – 005	FR-DA-SME-03, 06, 07, 08	
REQ-SME-SCORE-001 – 005	FR-DA-SCO-01, 02, 04	
REQ-SME-ONB-001 – 005	FR-DA-ONB-01 – 03, 05	
REQ-SME-PRICE-001 – 005	FR-DA-PKG-01 – 04	Packs are Solution Pack variants.
REQ-SME-UX-001 – 005	FR-DA-ONB-03, 04, NFR-DA-UX-02, 03	WCAG 2.1 AA and mobile use already required in Section 5.5.

Source family	Requirements in this standard	Note
REQ-SME-GOV-001 – 005	FR-DA-SME-08	
REQ-VS-ACT-001 – 003	FR-DA-SME-02, FR-DA-VRT-07, 08, Section 3.14	Vertical processes first, SME processes second.
REQ-VS-SCORE-001 – 003	FR-DA-SCO-03	
REQ-VS-CHK-001 – 003	FR-DA-CHK-01 – 07, FR-DA-GTE-05, 06	Checklists right-sized per vertical and track.
REQ-VS-GATE-001 – 003	FR-DA-GTE-01, 02, 04, 06	
REQ-VS-PKG-001 – 003	FR-DA-PKG-01 – 04	
REQ-NFR-PERF, SCALE	NFR-DA-PERF-06 – 08, NFR-DA-SCALE-04 – 05	Production targets; existing reference targets unchanged.
REQ-NFR-SEC, REL	NFR-DA-SEC-14 – 16, NFR-DA-REL-06	Audit and tenant isolation already covered (Section 4.11, NFR-DA-SEC-11).
REQ-NFR-UX, LOC	NFR-DA-UX-09 – 10, NFR-DA-I18N-02	RTL already covered (NFR-DA-I18N-01).
REQ-NFR-COMP, MNT; Part 7 governance	NFR-DA-COMP-05 – 06, NFR-DA-MAINT-08	

Product-specific material — the content of each vertical instance, the phase-gate vocabulary of each application and its demonstration data — stays in the documentation of that application.

Appendix F — Revision 1.4 Change Log

Revision 1.4 is additive: every earlier requirement keeps its ID and wording, and new requirements take the next free number in their section or a new family.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.5, 1.6	Version 1.4	Version raised to 1.4; purpose and overview describe the revision; ISO/IEC/IEEE 29148 and ReqIF added to the references.
1.4 Definitions	7 terms added	Standard, Standard Structure, Requirement, Requirement Type, Applicability, Statement of Applicability, Compliance Status.
2.3 Product Functions	1 function added	Standards and requirements management with applicability and compliance.
3.15 (new)	Architecture	Three catalogs, one record per requirement, inherited applicability, requirement links, ownership levels, licensed text.
4.3 Configuration Management	FR-DA-CFG-15	Compliance & Security Standards modules are Standards Catalog entries.
4.5 Knowledge Base	FR-DA-KB-05	Catalogs indexed for AI grounding and search.
4.6 Governance	FR-DA-GOV-09	Business Rules, Controls and Risks reference catalog requirements.
4.23 Checklist Library	FR-DA-CHK-08	Checklist items and gate criteria reference requirements and record evidence.
4.27 Vertical Compliance	FR-DA-VCP-04	Vertical standards held in the catalog and proposed on vertical activation.
4.36 Standards Catalog	FR-DA-STD-01 – 07 (new)	Internal and external standards with CRUD; structure from part and chapter to requirement; lifecycle and editions; import and export; AI draft; sharing and permissions; browsing.
4.37 Requirements Catalog	FR-DA-RQM-01 – 08 (new)	Typed requirements with CRUD; configurable types; one record reused; links between requirements; where-used; lifecycle and versions; filters, bulk edit, exchange; AI suggestions.
4.38 Applicability & Compliance	FR-DA-APL-01 – 07 (new)	Apply to Organization and/or project; inheritance and tailoring; defaults from templates and verticals; assessments with evidence; Statement of Applicability and matrix; alerts; audit.
5.1, 5.4, 5.9, 5.10	NFR-DA-PERF-09, SCALE-06, COMPAT-04, COMP-07 – 08	Import and matrix performance; catalog scale; CSV, XLSX and ReqIF round-trip; licensed text; requirement quality per ISO/IEC/IEEE 29148.
6 Data Model	7 entities	Standard (catalog), StandardNode, Requirement, RequirementType, RequirementLink, Applicability, ComplianceAssessment.
7.1, 7.2 Interfaces	Clarified	New screens and endpoint families for standards, requirements, applicability and compliance.
Appendix A	1 row	Conformance of the 1.4 additions in CortexPLM.